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Application of the Tree Algorithms to Analyze Factors Affecting Startup Success and Assessing their Accuracy

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Honorary Statement

I hereby declare that I have prepared my bachelor thesis on "*Application of the Tree Algorithms to Analyze Factors Affecting Startup Success and Assessing their Accuracy*" independently and I have identified and listed all the literature and other sources used in the attached bibliography.

Prague, 20.1.2025

.....

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Abstract

This thesis investigates the application of tree-based algorithms to analyze factors influencing startup success and evaluates their accuracy. The thesis employs three types of algorithms: Random Forest, Gradient Boosting, and simple regression tree. By applying these algorithms to a startup dataset, the research aims to determine their suitability for analyzing factors that may impact a startup's success, defined as the total funding raised by a company in U.S. dollars. The findings indicate that tree-based algorithms can serve as practical supplementary tools for decision-making processes by identifying key factors contributing to startup success. However, the study also highlights certain limitations of these methods, which are discussed in detail to provide a balanced perspective on their application.

Keywords

The startup, Decision tree, Random Forest, Gradient Boosting, Success factors, Predictive analysis, Decision-making support, Data-driven insights, Regression tree

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1. Introduction

Evaluating and predicting the success of startups is a key topic for both investors looking for opportunities with high growth potential and for startup founders themselves, who need to understand the factors affecting their success. Startups operate in an environment of high uncertainty and limited resources, often requiring strategic decisions based on available data. Identifying the key factors that influence success can help startups focus on those that increase the likelihood of their growth and stability.

In this thesis, I focus on using decision regression trees as a flexible and interpretable tool for analyzing data and identifying factors influencing their success. Decision trees are known for their ability to model complex and non-linear relationships between financial, geographic, or otherwise variables. The ability of decision trees to identify and visualize important distribution nodes makes it easy to understand how different factors affect the target variable.

Using decision trees to evaluate startups can have several significant benefits. This approach allows founders and investors to understand better which specific factors contribute most to their potential success, and the analysis results can help them increase their chances of growth. Startups represent a unique segment of the business environment, attracting investors because of their high potential for growth and innovation. Unlike established companies, they operate in a dynamic and rapidly changing environment but have a significantly higher risk of failure. Only 1 in 3 startups survive the first 3 years of existence (Santisteban & Mauricio, 2017), and according to other sources, the mortality rate is as high as 90% (Admin, n.d.; Patel, n.d.). According to CB Insights, the top reasons for failure include (*Why Startups Fail: Top 12 Reasons* | CB Insights, n.d.) :

- Running out of cash/failed to raise new capital
- No market needs
- Outperformed by competitors
- Flawed business model
- Regulatory/legal challenges
- Pricing/cost issues
- Not the right team
- Product mistimed
- Poor product/poor product
- Pivot gone bad

The definition of a startup is not clear. According to Santisteban and Mauricio, a startup can be defined as a new and temporary company with a business model based on innovation and technology. These companies have the potential for rapid growth and scalability and are recognized by governments worldwide for their contributions to economic stability, growth, and job creation (Santisteban & Mauricio, 2017).

Nevertheless, the high risk of the investment is outweighed by the potential gain. Startups are important organizations that play an essential role in developing economies and can be described as firms in the early stages of development. Statistically, the economic value of startups is very significant - in 2019, the estimated value of all startups globally was \$2.8 trillion, a 20% increase from 2017 (Spiegel et al., 2016).

It is important to analyze the factors that lead to startup failure, as the high failure rate of these firms can have significant economic consequences. Research shows that startups often face specific challenges, such as limited financial resources, unclear growth management, or inappropriate business strategies. Understanding these causes can not only help founders better prepare for obstacles but also provide investors and policymakers with valuable information to foster more effective business environments. Thus, the clear identification of these factors is key to preventing unnecessary startup failures and can help increase startups' chances of sustainability and growth.

2. The Researched Question

Startups are engines of economic growth and innovation but face a significant risk of failure. They significantly impact the likelihood of success but also face high failure rates. Statistics show that up to 90% of startups fail, and approximately 10% fail in their first year. Whether the cause is poor strategic planning or poor market understanding, analytics can help reduce the risk of this happening (Genome, n.d.).

What method should be used for analysis, and what success rate is expected from them? For example, the well-known and widely used SWOT analysis increases the chance of improving strategic decision-making by 15-25%, according to the Harvard Business Review (Yoffie et al., 2019).

Benchmarking plays an important role, allowing startups to compare their strategies and performance with competitors. Startups that use this method show up to 30% better performance on key metrics such as revenue growth or customer acquisition (*The 2020 Global CVC Report*, n.d.). Studies show that it can also be helpful to use the Lean Startup methodology, which offers a higher chance of survival than those that rely solely on traditional methods (*The Lean Startup | Methodology*, n.d.).

The subject of this thesis is whether the application of machine learning models and specifically tree algorithms, can provide sufficient predictive accuracy and whether it can reliably point to the factors that influence the success or failure of startups the most.

The success, and therefore the explanatory variable, is defined in this thesis as the total capital raised by the startup in USD.

The decision to use total capital as an explanatory variable is based on several key reasons. Capital raised is one of the most important measures of a startup's success because it directly affects its ability to grow, innovate, and sustain in a competitive market. This thesis aims not to determine the absolute amount that can be considered a success; this is highly individual, but to investigate whether using these methods for prediction and the limitations associated with their application is appropriate.

3. Literature Review

So, what is behind such a high number of failures, and what factors should be monitored in order for a startup to maximize its profits and reduce the risk of being classified as a failure? Many academic papers and research have tried to answer this question recently.

When analyzing datasets with publicly available information on firms seeking investment, regardless of the analytical technique used, we can focus on the factors that influence the failure and success of firms in the market and use these insights. Although the relevance of variables cannot be guaranteed until models are built, it can be supported by findings from systematic literature analyses focusing on factors leading to firm success or failure.

A systematic analysis of the factors essential for the success of information technology startups has been carried out, for example, by Santisteban and Mauricio. Their work examined a total of 1013 papers, from which 74 studies were selected for deeper analysis (Santisteban & Mauricio, 2017). In contrast, another systematic analysis (Noui & Dehane, 2023) focused on the analysis of factors leading to failure. It concluded the identification of 65 factors from a total of 353 post-mortem reports from the CB Insight Tech Market Intelligence Platform towards failure analysis was also undertaken by Akter and Iqbal (2020).

Santisteban and Mauricio defined a total of 21 factors important for startup success, divided into three categories. These include:

- general characteristics of companies and their founders
 - age of the organization
 - Size
 - venture capital
 - academic experience
 - Location
 - age of the entrepreneur
 - gender and so on
- Organizational factors, for example:
 - Location of clients
 - Investor locations

The influence of factors in the organizational category has been confirmed by other studies as central to the survival and further development of firms (Hormiga et al., 2011; Lussier, 2015).

The individual category included all factors related to human capital, i.e., the qualities of both the leader and the teams, and the last category of external factors included all factors related to the environment influenced by rivalry, pressure to innovate, and constant process changes.

At the same time, the authors of the studies acknowledge that undoubtedly important parameters, such as leadership skills, the entrepreneur's initial motivation, or the business plan, are sometimes difficult to measure or not publicly available.

However, some studies point out that failure in a particular area (a negative factor) may not permanently significantly harm a business if, for example, it has sufficient financial resources and the startup can afford to incorporate customer requirements more easily or withstand some failures (van Gelderen et al., 2005).

A systematic study by Noui and Denahe addressing just the sources of failure, which can sometimes be the same factors as success only viewed in reverse, cited the person of the entrepreneur as the leading cause of failure and used as a source of information not only academic papers but also analysis of post-mortem reports, a total of 353 compiled by CB Insights Tech Market Intelligence Platform(2023). Compared to other studies, they categorized the factors into a total of 8 dimensions, identified from academic articles, and then searched for them in the reports. Their definitions, along with examples of the factors included in them, can be found in Table 1. In addition to expected variables such as lack of experience or poor management, the study identifies a dangerous tendency to attribute a company's successes to internal and personal factors and its failures to external and situational factors.

Moreover, when the two types of failures meet, seeing problems only in external circumstances may be wrong. Microeconomic factors such as interest rates and unemployment rates, among others, can play a significant role in this sense, and according to Everett and Watson, these are positively correlated with an estimated failure rate of between 30 and 50%(1998).

Table 1-Dimension Definitions

DIMENSION	DEFINITION	EXAMPLES
ENTREPRENEURS	Founder's personal characteristic	Lack of enthusiasm, low educational level, Fear of failure
MANAGEMENT AND ORGANIZATION	Factors of organizational behavior, management, leadership, structure	Wrong pricing, poor marketing, lousy team, and time management
PRODUCT	Product/service characteristics, Performance	Wrong timing, bad quality, too high expectations
STARTUP		Bad location, Staff turnover
FINANCE	Factors related to financial and investor failure	Lack of capital cost issues, bad money management
CONSUMER/USER	Customer behavior, reactions, and everything related to it	Ignore customers, high acquisition cost, unloyalty
ECOSYSTEM	External factors: macro variables and competitors mainly	Law and regulation, Strong competition, taxation
ENTREPRENEURIAL PROCESS AND BUSINESS MODEL	Business model mainly	Bad business model, lack of market study

Source: Noui and Denahe (2023)

These factors identified in the theoretical literature were compared with those reported in industry reports. They measured the occurrence of these factors as a percentage of their representation in terms of a particular perspective. Noui and Denahe also include the probability of failure in the phase in which the startup is. They summarized the divisions of many different papers, for example, Bocken(2015) and proposed a division into these 4 phases:

Figure 1-Stages of startup development



Source: Bocken(2015)

By measuring the prevalence of these factors in post-mortem reports, they show that startups fail most often in the early-growth phase, a complete 50.71%. These are companies under 5 years old with 10 to 100 employees and a funding size of less than \$50 million.

Following in the tail are then scale-ups¹ with 100-1000 employees, existing for less than 10 years and total funding under 250 million USD. Start-up phase firms then fail in 17% of cases. In contrast, the Mature or Idea phase failed only minimally.

The gender perspective also plays an interesting role, showing that most start-ups are founded by men. However, when examining the factors influencing failure, it is important to note that this unequal representation can lead to significant cognitive biases in the results. The number of start-ups founded by women is significantly lower, which affects the interpretation of the failure data. Indeed, the baseline varies radically: for example, a study from Startup Genome, which included a survey of over 5,000 respondents in 67 global ecosystems, found that only 31% of startups have a female founder, while on average, only 15% of startups are founded by women. Therefore, when analyzing factors of failure, it is necessary to take into account who founded the startup. At the same time, gender aspects need to be examined together with the geographical location of the company, as most women found startups in North America and Oceania, while the least in the MENA region (Stefanuto, 2023).

However, any categorization of factors into different groups and assessment of their impact on companies is complicated by the specificities of many sectors. For example, regulatory criteria can vary widely and often evolve precisely in response to the behavior of firms in the market across years, particularly in relatively new and fast-growing sectors. Typical examples are so-called platform startups. These types of businesses are more often having to deal with, among other things, stricter laws to protect user data, which can potentially put their business model at significant risk. For example, businesses face significant financial penalties for violating data protection rules, as well as loss of customer trust.

Another study has identified failure to adapt and cope with the volume of legislative requirements as a significant factor of failure, especially for small start-ups that do not have the human or financial resources to cope with such a burden (Akter & Iqbal, 2020). Also, unsuspected legal liability for customer harm caused by the product's mere existence can threaten a start-up's existence. Successful lawsuits against various social networking sites for allowing the dissemination of pornographic, violent, and otherwise illegal material have already taken place in several states (Khajeheian, 2020). Other specific failure factors then included failure to engage developers, protect monetization opportunities, and poor pricing, ignoring competitors (Yoffie et al., 2019) (Davidsson & Gordon, 2012).

¹ Scale-up is the stage following a start-up: a relatively young, rapidly emerging, innovation-driven company, that's experienced a high level of growth over the past three years (Forbes).

3.1. Geographical Categorical Variables

However, other research emphasizes the impact of geographic variables on firm value or other related economic indicators.

Berre explored the use of the tree algorithms in combination with OLS or DFC methods to predict the firms' pre-money valuation. He compared the variables included in the DCF model and their predictive power with the performance of categorical variables, specifically geographic variables indicating the location of investors and firms. The result was very different startup valuations in different regions in practice despite comparable firm revenues or various other variables identifying market conditions.

The findings of the research confirm that the tree algorithms can be used very effectively to estimate the value of firms prior to investment and can be sufficient in situations of data unavailability or secrecy. Many of these algorithms are so-called externally information dense. In terms of the power of specific variables, a country-level variable can replace, for example, cash-on-market, and a city-level variable can replace, for example, municipal GDP (2024).

Baptista examines the effect of geographic location on the diffusion of innovation underlying the startup effect. The study demonstrates that geographic factors play a key role in the diffusion of technological innovation, particularly through local learning effects in geographic clusters, where proximity to early adopters and similar firms encourages collaboration and facilitates access to new technologies through contacts and knowledge sharing. (2000). These effects are particularly strong at the early stages of the diffusion of innovation, and their influence on firm success is confirmed by Porter, who argues that national prosperity is not a legacy but the result of innovation and development that fosters a competitive environment(1990). Successful firms adapt to pressures and challenges, which leads to sustaining their competitive advantage in the global marketplace. The development of geographic areas is also linked to the success and prior success of firms by Pe'er and Keil(2013).

4. Methodology

Decision tree-based regression models were chosen to effectively capture the complex and non-linear relationships in the data.

These machine learning algorithms are non-parametric supervised algorithms used for regression and classification tasks (*What Is a Decision Tree?* | IBM, n.d.). Their exact date of origin is difficult to determine; however, statisticians Leo Breiman and Charles Joel Stone of the University of Berkley are considered to be the authors of tree algorithms, who, along with Jerome H. Friedman and Richard Olshen of Stanford University, began developing the Classification and Regression Tree Algorithm (CART) in 1977. The first official publication with CART software was published in 1984 (Foote, 2021). Even today, CART methods are often used in data analysis, but their implementation is mainly in programs like R-studio or Python. Decision trees in Python programs are based on the CART methodology but have some different features and modifications in the implementation. For example, they have built-in parameters and hyperparameters for managing tree complexity or pruning complex trees, offering more options for scaling models to large datasets.

There are two types of decision trees, namely classification and regression tasks. Classification ones are used for a categorical target variable, and the data is divided into different classes or categories. Metrics like the Gini index or entropy are used as the criteria for data partitioning to find the optimal points of the partition. Then, for a continuous target variable, there is a regression variant, which is used in this thesis, and for the distribution of decision nodes, it uses the sum of squares error minimization method to find the optimal distribution and, therefore, the least variability within the resulting groups.

5. Tree Algorithms: Theory

Decision trees are machine learning methods based on regression and classification. The methods involve stratifying or segmenting the predictor space into several simple regions. The name decision tree methods come from the arrangement of splitting rules, which are applied to divide the predictor space, which can be repressed in the form of a tree. The advantage of tree algorithms is their ease of interpretation. However, they have poorer performance in terms of prediction accuracy or the ability of the model to explain variability in the target variable. The solutions to this shortcoming are ensemble methods such as random forest and gradient boosting, which are based on different principles but have in common the use of a large number of separate decision trees that are combined into a single prediction, often yielding substantial improvements in prediction, but at the cost of poorer interpretability. All models based on decision trees exist in both regression and classification versions. Regression models are used in this thesis, so we specifically focus on them.

5.1. Decision Trees

A simple regression decision tree starts at the so-called root node with no predecessor. The root node is then divided into left and right branches based on the selected attribute and its value. The node is then divided into a left branch, which corresponds to $X_j < t_k$, and a right branch, which contains values corresponding to $X_j \geq t_k$. Trees are usually further subdivided into other nodes, which have further descendants. Nodes that subdivide are called internal nodes. Those that do not subdivide are then called terminal (leaf) nodes (James et al., 2023, p. 331).

5.1.1. The Process of Decision Tree Formation

The formation of the decision tree can be divided into three parts:

a) Predictor space partitioning

The decision tree progressively divides the predictor space into smaller regions using decision rules that are based on the values of each predictor.

b) Regions (Stratification)

After several subdivisions, regions (R_1, R_2 to R_j) are formed, which correspond to different paths through the tree. Each leaf node corresponds to a region in the space of predictors (X_1, X_2 to X_p).

c) Predictions in the region

In each region, the tree predicts the value of the target variable based on the data that falls within that region. So, for example, if given observation $X=x$, then if it falls in region R_1 , then we will predict a value of 30, and if it falls in region R_2 , then the prediction value for that observation will be 40. The division of nodes into other nodes continues until a stopping criterion is reached.

In the Tree Pruning section, we discuss stopping criteria for decision trees (pruning) more.

The feature based on which the root node splits (predictor), based on which the region is formed, is selected based on maximizing the quality of the split. For regression decision trees, this means splitting the values of the variables in an attempt to achieve the best possible value of the error metric (James et al., 2023, p. 334)

The goal of the decision tree is to find regions that minimize the RSS mean quadratic error. Since it is practically infeasible for the tree to compute all possible distributions in advance and only then partition the data into nodes. For this reason, a top-down approach, "recursive binary splitting," is applied *when the best split is made at a particular step, rather than looking ahead and picking a better split that will lead to a better split in the future steps.*

$$RSS = \sum_{j=1}^J \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

Equation 1-Residual Sum of Squares

Thus, the tree uses the root mean square error to partition the values into regions by comparing the difference between the actual values of the target variable y_i and the predicted values. The goal is to partition the data to be as homogeneous as possible within each region, thus minimizing the error. However, a problem arises as soon as we would like to compare the ability of node distributions in different regions or the performance of different models since this metric is absolute, and each distribution operates with a different number of samples.

Therefore, at each node and for the whole model, we compute a normalized metric, MSE, the mean of the error, which comparisons between nodes and models with different numbers of samples allow and are commented on in the section on metrics evaluating model performance (James et al., 2023, p. 334).

5.2. Tree Pruning

The trees terminate the division based on reaching specific criteria values. These are either present to allow the tree to split almost indefinitely, for example, allowing the absolute minimum of samples in leaf nodes, or they do not limit the maximum depth to which the tree can grow.

The problem, however, is that overly complex trees are prone to overfitting, where the algorithm learns every subtle nuance in the data and then generalizes poorly on test data it has not yet seen. Conversely, a smaller tree may not only be more interpretable but also runs the risk of underfitting, where the model is not complex enough to capture complex relationships in the data. There are two types of pruning, namely pre- and post-pruning. Post-pruning is based on growing an unrestricted tree, and then it is pruned back in order to get a subtree.

However, this approach is very computationally intensive and, therefore, is not used even for comparison with pre-pruning. The latter is a way of growing a tree that is already constrained by setting hyperparameters to specific values, such as the maximum depth of the tree, before the learning process starts. The hyperparameters are not original components of the CART tree algorithms mentioned earlier, and to find a complete list of them, one can visit the official documentation of the scikit-learn library. For the regression decision tree, the values of the following hyperparameters are modified:

Table 2-Importance of hyperparameters in pre-pruning decision trees

Source: Scikit-learn.org

Hyperparameter name	Meaning of the Hyperparameter
max_depth	The maximum depth of the tree. If None, nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples.
max_samples_leaf	The minimum number of samples required to be at a leaf node. A split point at any depth will only be considered if it leaves at least min_samples_leaf training samples in each of the left and right branches.
max_samples_split	The minimum number of samples required to split an internal node
random_state	Controls the randomness of the estimator. The features are always randomly permuted at each split. To obtain a deterministic behavior during fitting, random_state must be fixed to an integer.

Hyperparameters² specific to ensemble methods are defined later.

5.3. Ensemble Methods

A simple decision tree is generally considered to be a model that achieves only average performance and is a so-called *weak learner*. Ensemble methods use the power of multiple trees, whose results then combine to form a single strong prediction. The methods used in this thesis are random forest and gradient boosting, which are based on bagging and boosting methods.

Bagging is based on the bootstrap method. It is a repeated sampling from one dataset. Bagging parallels a number of independent models and trains them on these different and random subsets of the data. The results of each independent model are then averaged for the regression models. The goal is to reduce variance, improve model robustness, and reduce the risk of overfitting (James et al., 2023, pp. 343-344).

Compared to bagging, *boosting* works on the principle of sequential training. Each model is trained based on the results of the previous model and focuses on the mistakes made by that model. At the same time, each model is given a weight that determines its importance in the overall result of the whole large model.

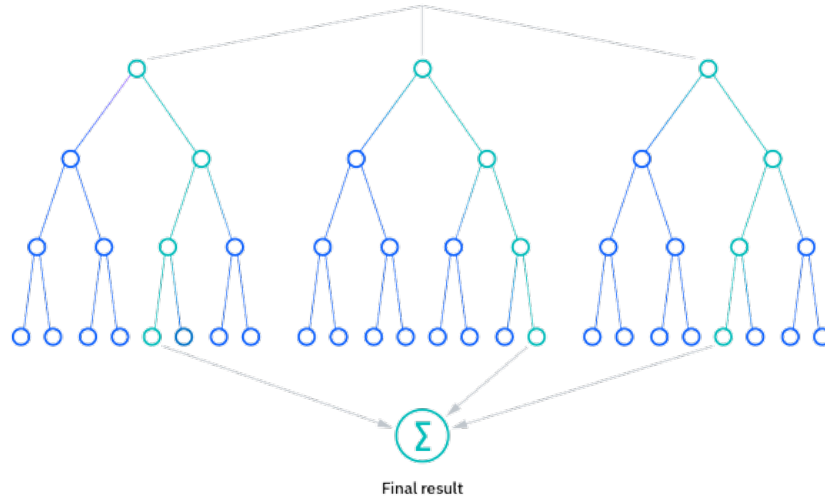
5.3.1. Random Forest

This algorithm compiles a number of models separately, the result of which is averaged. Unlike bagging, however, it differs in the number of predictors that are used in model building. Each time a node might be split, only a subset of m predictors is randomly selected from the total number of p predictors. Usually, m is approximately equal to $\sqrt{p}$.

This is preferable to bagging. This is because if there are dominant predictors for splitting, nodes are always selected from the complete set of predictors; the trees are strongly correlated. At the same time, the random forest reduces the correlation. Generally, good results are obtained with a more significant number of trees, but the computational effort increases, which may pose some technical limitations (James et al., 2023, pp. 346-347).

² Hyperparameters are values that are set before the training process begins and are not estimated from the data, parameters : model learns them automatically during the training process(Oliwa, 2024).

Figure 2-The principle of the random forest algorithm



Source: IBM

5.3.2. Gradient Boosting

Gradient Boosting (GBR) is based on the principle of updating the model prediction using a weighted combination of the previous tree and the previous model, and each model is trained by targeting the errors of the previous one (*Gradient Boosting in ML - GeeksforGeeks*, n.d.). However, each tree is trained on the current residuals.

$$r_i = y_i - \hat{y}_i$$

Equation 2-Residuals

The process of creating a GBR starts with determining a baseline value, which in regression is usually the average of all outputs. If we predict the total amount of capital raised by startups, and it is \$10 million, the model will start at this point for all startups.

$$f_o(x) = \bar{y}$$

Equation 3-Initialization of the model

The model then determines how much its predictions differ from the actual values, and the model is trained to predict these residuals. The model then updates the prediction by adding a new tree. The new model is corrected by a so-called shrinkage factor (η), which determines how fast the model learns. Slowing down improves generalization and reduces the risk of overlearning. Higher values then cause the model to learn faster. In python, the shrinkage factor is called the learning rate (James et al., 2023, pp. 347-349).

$$f_b(x) = f_{b-1}(x) + \eta \cdot f_b(x)$$

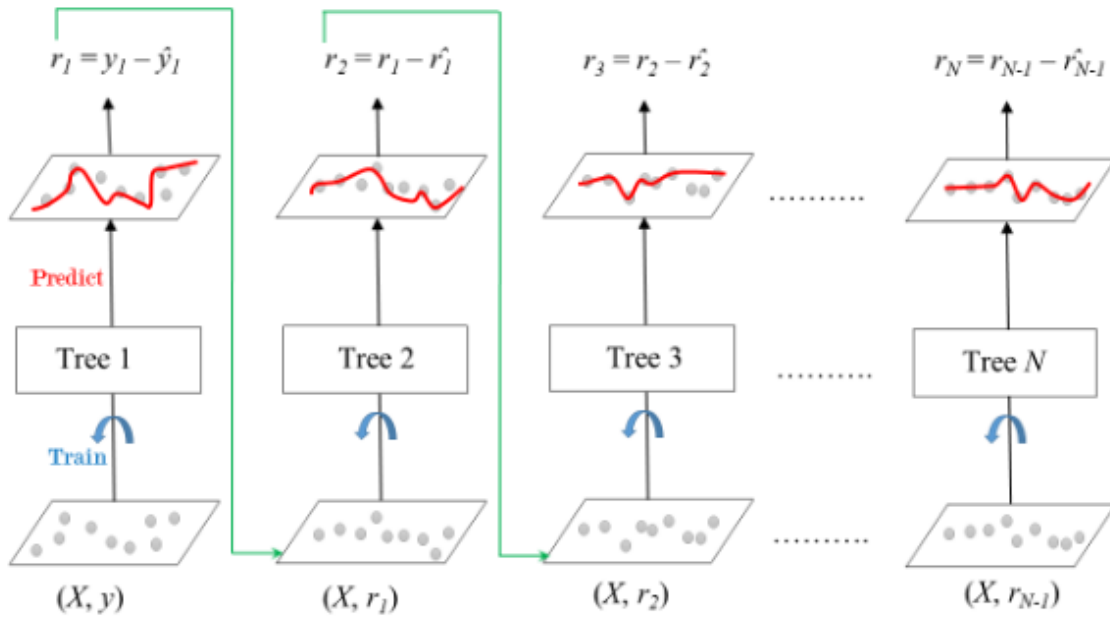
Equation 4-Updating the Gradient Boosting model

For example, if the first startup raises \$9.8 million and the second model adds a \$0.1 million correction, the prediction for the first startup would be \$9.9 million. This process is then repeated for the other trees. The following formula can define the final model:

$$f(x) = b = \sum_{b=1}^B \eta \cdot f_b(x)$$

Equation 5-Final Gradient Boosting model

Figure 3-Principle of Gradient Boosting



Source: (Gradient Boosting in ML - GeeksforGeeks, n.d.)

5.4. Hyperparameters in Ensemble Methods

In addition to the hyperparameters already mentioned for the simple regression tree, there are also specific parameters for ensemble methods. One of them is `n_estimators`, i.e., the number of trees in the method, which significantly affects the accuracy and quality of the model. Then, for the gradient boosting model, the learning rate is already mentioned for GBR.

For random forest, another important hyperparameter is `max_features`, whose correct setting will avoid the use of all variables, thus increasing the diversity of the trees. In addition to the square root of the total number of variables (`sqrt`) can be set, and in this thesis the value `log2` is used to quickly reduce the number of variables selected for each decision node. For example, for 1000 variables, it would select approximately ten (`RandomForestClassifier`, n.d.).

5.5. Tree Algorithms versus Classical Regression

Linear regression is a popular method, but it is not used in this paper. However, the advantage of decision trees over this method is the ability to model non-linear relationships between predictors. Moreover, in linear regression, the relationships between predictors must be explicitly included in the model of human interactions ($x_1 \cdot x_2$ and so on).

Tree-based algorithms also have the advantage of the model's ability to produce a local prediction. In contrast, linear regression produces a single global prediction valid for the entire dataset (James et al., 2023, p. 341).

6. Verification of Model Quality

The performance of regression models is standardly verified using several methods, usually combined to show how well the model captures the relationships in the data and its ability to predict on previously unknown data. This subsection describes the methods and metrics commonly used to assess the quality of regression models. Specifically, the coefficient of determination (R^2), mean squared error (MSE), mean absolute error (MAE), and root mean square error (RMSE).

6.1. MSE

Mean Square Error (MSE) is a standard loss function quantifying the mean square error between the actual and predicted values. A lower value of the index indicates higher model accuracy, while a higher value indicates more significant differences between the actual and predicted values and, therefore, poorer model performance. The quadratic approach magnifies the impact of more considerable differences by sensitizing the indicator to outliers and outliers (*Mean Squared Error*, n.d.).

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Equation 6-Mean Squared Error

Where:

- N is the number of observations in the dataset
- y_i are the actual values
- $\hat{y}_i$ are the predicted values
- $(y_i - \hat{y}_i)^2$ quadratic error
 - The square ensures that all errors are positive and emphasizes more significant errors as they increase quadratically

6.2. MAE and RMSE³

Mean absolute error is one of the most commonly used metrics in regression models, measuring the average of the absolute differences between predicted and actual values, as well as overall values in a dataset (Ahmed, 2023).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Equation 7-Mean Absolute Error

Where:

- n is the number of observations in the dataset
- y_i are the actual values
- $\hat{y}_i$ are the predicted values
- $|y_i - \hat{y}_i|$ is the absolute error between the actual and predicted value for the i th observation

Compared to RMSE and MSE, MAE is robust to outliers. The indicator is expressed in the same units as the target variable, and all error metrics are weighted equally.

6.3. RMSE

Root mean squared error is also a metric observing the difference between predicted and actual values, penalizing larger errors due to the quadratic term in the formula (Team, n.d.). The formula of the metric is:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Equation 8-Root mean squared error

Where:

- n is the number of observations in the dataset
- y_i are the actual values
- $\hat{y}_i$ are the predicted values
- $(y_i - \hat{y}_i)^2$ is the square of the error between the actual and predicted value for the i th observation

³ MAE and RMSE are considered only supplementary indicators

RMSE can help penalize significant errors in the model. Unlike MSE, which loses direct interpretation by quadratic errors, RMSE returns the error in the same units as the target variable.

6.4. R^2

The coefficient of determination is used to explain the variability of the dependent variable in the regression model or how well the dependent variables explain the variability. Thus, R^2 provides a measure of how well samples on future data will be predicted by the model, with the relationship again based on the difference between the actual and predicted data. The indicator takes values between 0 and 1 but can take negative values if the model is very poor. It is also important to note that as more and more variables are added, the indicator may increase despite the fact that these variables are insignificant to the model (*R-Squared*, n.d.) .

The formula R^2 is as follows:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Equation 9-Coefficient of determination

Where:

- SS_{res} (sum of squares of residuals) is the sum of squares of residuals, i.e., the differences between predicted and actual values.
- SS_{tot} (total sum of squares) is the total sum of the squares of the differences between the actual values and their average value.

Values:

- A value of 0 means that the model variables do not explain any variability in the dependent variable
- A value of 1 means that the model variables perfectly explain the variability of the dependent variable

However, when assessing the quality and reliability of regression models, it is also important to use metrics that provide information not only about the accuracy of the model but also about its interpretability and ability to be generalized to new data. Two key techniques that are often used in this context are feature importance and cross-validation.

6.5. FMSE

FMSE or Friedman MSE is a loss function based on reducing the mean square error at individual nodes in the construction of trees in Gradient Boosting, so it is not used to compare the performance of individual models, and John H. Friedman described the concept.

The FMSE is calculated as the difference between the error in the parent node and the weighted average of the errors in the child nodes. The variable that contributes the most to reducing the MSE is then considered the key variable for the distribution of a given node. The metric also plays a key role in determining the importance of variables in Gradient Boosting, as it reflects their contribution to the error reduction of the model (Friedman, 2001).

$$L(y, f(x)) = (y - f(x))^2$$

Equation 10-Loss function

In the book *Classification and Regression Trees* by Brieman et. al(2017), a formula is then described for evaluating the quality of the distribution of nodes in the decision tree, which quantifies the improvement caused by splitting the data into two subsets L(left branch) and R(right branch), and thus allows to determine the variable that has the best split.

$$\Delta MSE = MSE_{parent} - \left(\frac{|L|}{|L| + |R|} MSE_L + \frac{|R|}{|L| + |R|} MSE_R \right)$$

Equation 11-Calculation of MSE change

Where:

- MSE_{parent} : mean squared error in the parent node before splitting
- MSE_L, MSE_R : left and right branch errors after splitting
- $|L|, |R|$: Number of data points in the left and right branch

6.6. Feature Importance

When building and evaluating machine learning models, it is crucial to determine which input variables affect predictions the most. To do this, so-called feature importance can be used. By analyzing the feature importance, we can obtain the influence in the predictions in the form of a percentage of all variables and determine whether they have a greater or lesser influence on the value of the target variable.

The method of calculating feature importance varies according to the type of algorithm and model. In classification models, feature importance is calculated based on impurity reduction, which means that the model looks at how a particular variable contributes to improving the purity of the distribution at the nodes of the tree. The Gini index or entropy is commonly used to measure impurity in classification models. In regression models, feature importance is calculated through variance reduction. That is, it looks at how each variable in a node contributes to reducing the variance (variance) of the values in the leaf nodes (*Feature Importance*, n.d.). The more a particular variable contributes to reducing variance, the more important it is considered. This approach allows us to express how much influence each variable has on the model's ability to predict values (Breiman, 2001) correctly.

In addition, in Random Forest models, there is another method of assessing the importance of variables known as Out-of-Bag (OOB) Error. OOB samples are data points that were not used to train a particular tree in the forest. The values of each variable in these samples are randomly shuffled to calculate the OOB error. If the accuracy of the prediction deteriorates significantly after shuffling a variable, it means that this variable has a significant effect on the model's decision-making and is considered important (Ronaghan, 2019).

6.7. K-Fold Cross Validation

Cross-validation is a method used to improve the accuracy of model estimation and to reduce the risk of overfitting. Its principle is to randomly divide the data into several k groups (so-called k-folds). The first fold acts as a validation set, and the remaining k-1 folds act as training (3.1. *Cross-Validation*, n.d.). Thus, it is a convenient addition to quality validation because it reduces the influence of a particular data split on the training and testing part of the data. At the same time, each set is used once for testing and k-1 times for training.

A metric evaluating the model performance, usually MSE, is also computed at each validation fold, i.e., overall MSE_1 to MSE_k . The k-fold CV estimate is then computed as the average of all the computed MSE values.

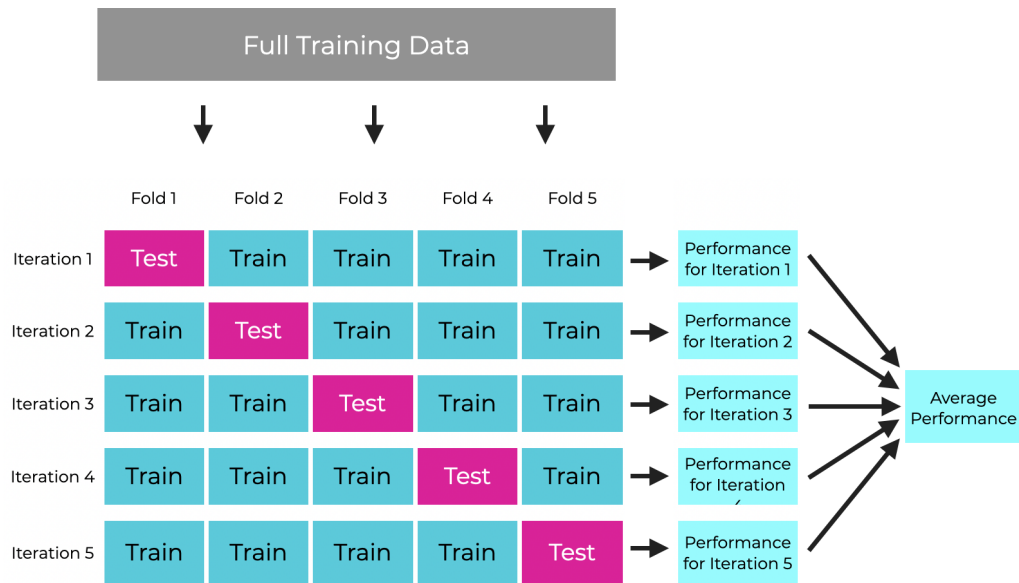
This reduces the influence of random outliers that can arise from a one-shot distribution of the data. The number of folds is then a matter of individual decision, and there is no mandatory standard, but usually splitting the dataset into 5 or 10 parts is used.

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i$$

Equation 12-Calculation of Cross-Validation metrics

For example, if a model performs well on training data but poorly on test subsets, this may indicate overfitting to specific patterns in the training set and a risk of poorer performance on unseen data (Shaikh, 2018).

Figure 4-Principle of K-fold Cross Validation



Source: sharp sight labs.com

6.8. Grid Search

Grid Search is a standard machine learning algorithm used to tune hyperparameters, i.e., parameters that we specify to the model ourselves, not the model learning them on its own. They can be preset in various machine learning libraries, but we can change their values ourselves. This process consists of searching through all possible combinations of hyperparameter values defined in advance to find the optimal model. However, it also has its limits. The method is computationally intensive, especially when dealing with a large number of hyperparameters, making it ideal for smaller datasets (*Grid Search*, n.d.).

7. Data

Table 3-Variables in the reduced dataset under study

Source: Crunchbase

Column name	Meaning of the columns
num_funding_rounds_organization	The number of investment rounds the company has gone through.
total_funding_usd_organization	Total capital raised by the company in US dollars.
investment_count_investors	Number of investments by the investor.
total_funding_usd_investors	Total capital raised by the investor in dollars.
investor_count_funding	Number of investors in the company in the investment round.
Presence_on_social_media_organization	Number of social networks the company is on (max 3)
Presence_on_social_media_investors	Number of social networks the investor is on (max3)
type_organization	Type of organization
roles_organization	Roles of organization
country_code_organization	Country code of the country of the organization
employee_count_organization	Number of employees in the organization
primary_role_organization	Roles of the company - e.g., Investor, etc.
type_investors	Type of investor-individual investor, company
investor_type	The distinction between the investor as an individual or an organization
is_lead_investor	Distinguishing whether or not it is the leading investor
type_funding	Funding type (always funding_round in bulk-export)
roles_investors	Roles associated with investor, e.g., company, investor
country_code_investors	Code of the country where the investor has its head office
investor_types_investors	Investor type, e.g., venture capital, micro_vc, etc.
type_investors	Type of financing (always an investment in bulk export)
country_code_funding	Country code associated with a funding round
investment_type	Type of investment, e.g., Seed, series, etc.
type_of_industry_organization	The distinction between the technological or non-technological sector in which the company operates
founded_on_organization_decade	The decade in which the company was founded
last_funding_on_organization_decade	The decade when the company's last round of financing took place
announced_on_funding_decade	The decade when the funding round was announced
founded_on_investors_decade	The decade when the investor was founded
city_organization	The city the company is located in
city_investors	The city the investor is located in
city_funding	The city that is associated with the funding round

The data source for this thesis is the Crunchbase database, and in the final sample, we have data on 12,024 companies, which are defined by the following variables in Table 3 (Data Dictionary, n.d.).

The data was created by combining four separate datasets, which were further edited and analyzed and were downloaded on 31.3.2023. *Funding rounds* contain data on individual rounds undertaken by companies, *investors* contain data on investors, *investments* contain data on investments, and information on organizations contains a set of *organizations*. The resulting dataset is an amalgamation of all the previous ones and has a total of 20,725 rows.

A statistical overview of the final sample is presented first, and the methods for adjusting it are described next. The explained variable *total_funding_usd_organization* in the Crunchbase data reflects the total amount of funding received by an organization across all funding rounds. These amounts are standardized to US dollars by converting them to USD based on the exchange rate in effect on the day the funding round was announced (Teare, 2021).

7.1. Statistical Summary

7.1.1. Categorical Variables

The final sample to which we will apply the regression decision tree algorithms consists of 18 variables divided into categorical and numerical variables for clarity. It is also important to see how many artificial variables will be generated later when applying the tree algorithms in the scikit-learn library.

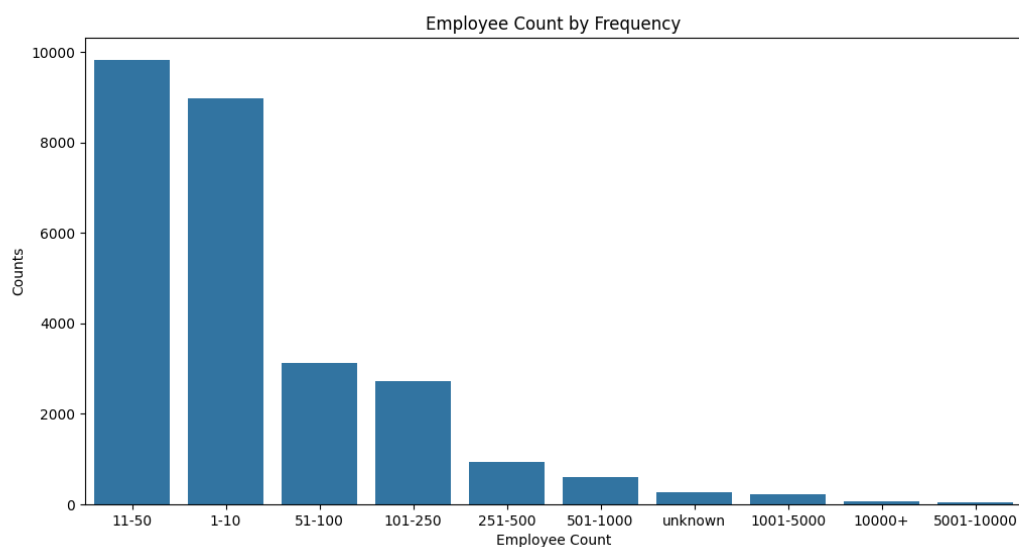
Table 4-Categorical variables in the final sample

Column name	count	unique	top	freq
roles_organization	20725	1	company	20725
country_code_organization	20725	2	US	19196
city_organization	20725	2	non-core-startup-city	13985
employee_count_organization	20725	10	11-50	7500
investor_type	20725	1	organization	20725
is_lead_investor	20725	3	undifferentiated	11487
roles_investors	20725	6	investor, company	18601
country_code_investors	20725	2	US	19562
city_investors	20725	2	non-core-startup-city	13894
investor_types_investors	20725	3	Financial Investors	12174
country_code_funding	20725	2	US	19196
city_funding	20725	2	non-core-startup-city	13985
investment_type	20725	8	seed	9463
type_of_industry_organization	20725	2	Technology	14664
founded_on_organization_decade	20725	7	2010-2020	16992
last_funding_on_organization_decade	20725	4	2020+	10721
announced_on_funding_decade	20725	4	2010-2020	13947
founded_on_investors_decade	20724	8	2010-2020	8779

Source: Crunchbase

The overview shows that companies based in the United States dominate, and only about 2,000 startups come from another country. A closer examination of what original countries are hiding under the value "non-US" reveals that it hides only Canada, resulting in a rather significant loss of geographic information, which is unintended but due to the deletion of blank values in other columns. Small firms between 11 and 50 employees are the most represented, in 36% of cases.

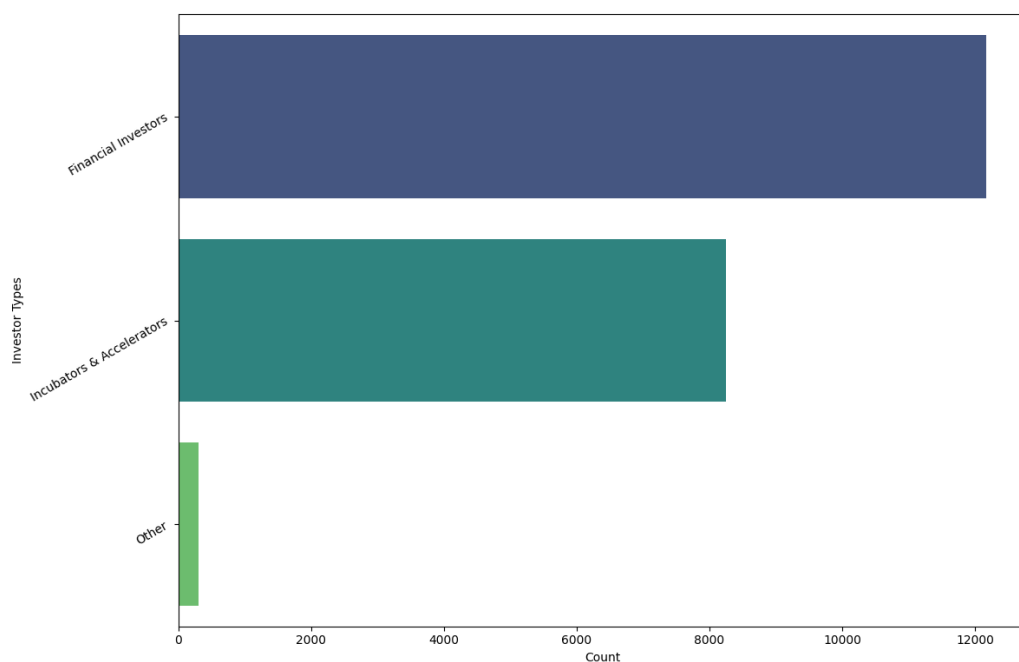
Figure 5- Number of employees in the firms represented in the final sample



Source: Based on data from Crunchbase

Overall, the share of firms with a maximum of a few hundred employees is predominant, which corresponds to the normal market situation for firms at the time of reaching Series C. The share of firms with a thousand or more employees is a minority in the final sample. A larger fraction of the firms are in the technology sector and are most often backed by an investor who is labeled a financial investor.

Figure 6- Distribution of types of investors in startups in the final sample



Source: Based on data from Crunchbase

The vast majority of firms were founded after 2010, and a significant proportion of firms received their last financing around 2020. The difference between the firm's founding date and the last financing so far is usually between zero and 10 years.

7.1.2. Numeric variables

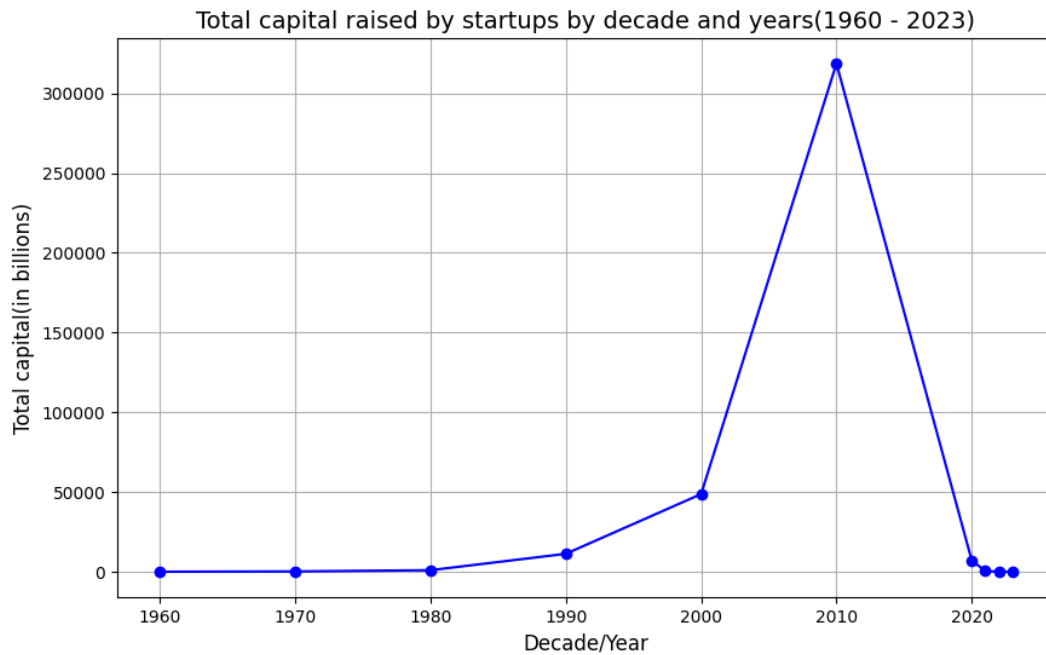
Table 5-Numeric variables in the final sample

Variable name	count	mean	std	min	x
num_funding_rounds_organization	20725	4,59	2,82	1	x
total_funding_usd_organization	20725	64289545,94	349525715,9	4907	x
investment_count_investors	20725	1463,94	2047,54	1	x
total_funding_usd_investors	20725	70378233,09	502159075,3	0	x
investor_count_funding	20725	5,16	4,75	1	x
Presence_on_social_media_organization	20725	2,41	0,9	0	x
Presence_on_social_media_investors	20725	2,63	0,66	0	x
Variable name	count	25 %	50 %	75 %	max
num_funding_rounds_organization	20725	3	4	6	26
total_funding_usd_organization	20725	2200000	12062250	48120000	15205232806
investment_count_investors	20725	95	257	1541	5425
total_funding_usd_investors	20725	0	500000	19678479	11000120000
investor_count_funding	20725	2	4	7	95
Presence_on_social_media_organization	20725	2	3	3	3
Presence_on_social_media_investors	20725	2	3	3	3

Source: Crunchbase

In the overview of the numerical variables, we see that, on average, one company has undergone four investment rounds, but we also find an outlier of 26 funding rounds. A maximum of 95 investors backed startups; however, this may be an outlier as 75% of companies were funded by up to 7 investors.

Figure 7- Distribution of total capital raised by startups

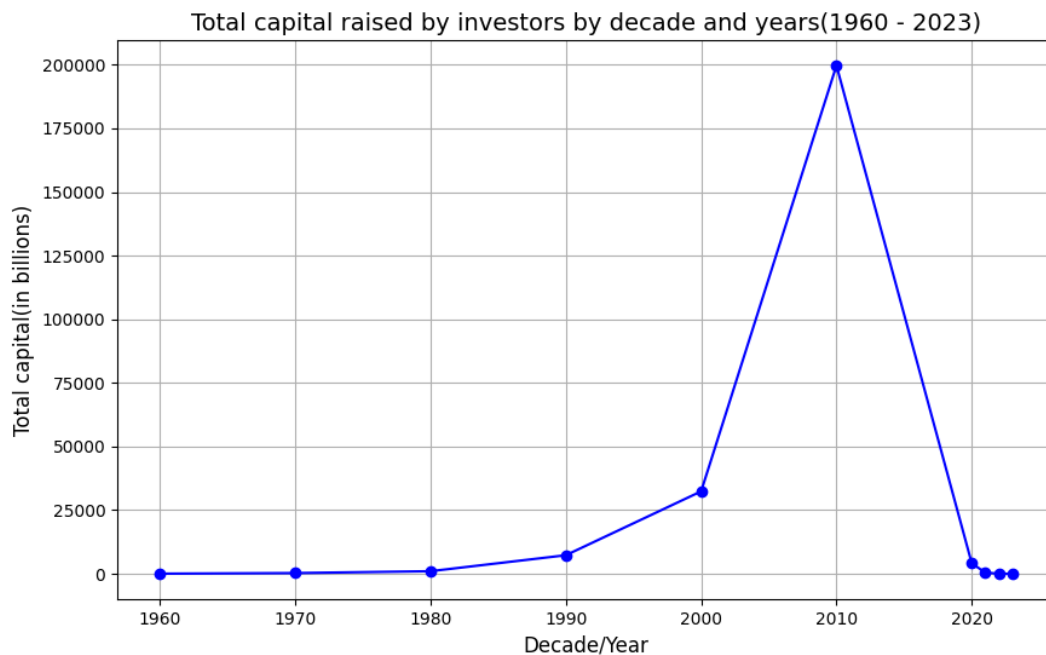


Source: Based on data from Crunchbase

The volume of startup capital peaked in 2010 and then declined until 2020. By 2020, it had reached almost zero. However, this is not where our data replicates the market as a whole. For example, according to statista.com (*Venture Capital - Worldwide | Statista Market Forecast*, n.d.), venture capital investment in the US (which makes up the majority of the resulting dataset) was \$345.4 billion in 2021, a record high, and \$242 billion and \$170 billion in the following years. However, this dataset contains a small fraction of all venture capital-funded firms, so it is impossible to relate global market developments to our sample.

At the same time, the values in our data are widely dispersed, as can be seen from the mean and standard deviation. Similarly, the amount of money raised by investors can be compared. However, the median number of investments made is only 284. The representation of different sizes of both investing and demanding investments is positive in terms of the algorithm's ability to learn from different values; outliers will then be solved by logarithmic transformation.

Figure 8- Total capital raised by investors



Source: Based on data from Crunchbase

Investors also peaked in terms of capital raised in 2010 and overall raised less capital in the years under review than companies raised. This may be due to missing values, which then caused the entire row to be deleted. There is also a discrepancy in the size of the dataset for investors and the dataset for organizations. The investor dataset was smaller, and so some investors are definitely not included in our data. Investors then made an average of 1,484 investments.

7.2. Method of Data Acquisition

Crunchbase obtains the data used in this work in several ways. First, investors and investment companies themselves regularly update information in their investment portfolios, for example, sharing details of the funding round a particular startup has gone through. However, other contributors include the companies themselves, who are looking for an investor. Finally, even Crunchbase itself uses its own data team to search for different companies on their own and can add or correct the data. The form of the data can vary according to the levels of access to the database. Crunchbase Pro, which includes a CSV export option and various advanced filters or enrichment of the data in the database with data from collaborating companies' databases. Crunchbase Enterprise also includes an API, regular database snapshots, and complete data exports (*Product Pricing*, n.d.).

This thesis uses a version of the Crunchbase API through which it was possible to download several datasets simultaneously through a company-provided key. These are structured datasets in the form of CSV files that can be linked using unique ID keys and always relate to a specific type of information. All files were linked together using Python's inner merge function

8. Dataset Preprocessing

Raw data is usually not ideal for subsequent application of machine learning algorithms and usually contains errors, incomplete records, or values that can negatively affect the accuracy and performance of the model. Therefore, this data must be transformed and cleaned before it can be used. This chapter explains the specific data modifications I have made and for what purpose. Data preparation includes but is not limited to filling in missing values, removing outliers, and transforming categorical and numerical variables, steps that have a significant impact on the effectiveness of subsequent analytical methods. The original dataset before the modifications had 712,616 rows and contained data on 158,674 organizations. Table 6 below shows the representation of variables in each dataset before linking and editing.

Table 6 - Variables in source datasets (before merging and editing)

Original dataset	Categorical variables	Numeric variables	Number of lines
Funding rounds	18	6	545 600
Organization	36	5	2 713 096
Investors	21	4	245 310
Investments	13	1	885 815
Total in dataset	88	16	712 616 ⁴

Source: Crunchbase

8.1. Solving Missing Values and Removing Columns

Missing values can sometimes be handled, but the decision tree algorithms of the scikit-learn⁵ library cannot handle them; therefore, they must be removed.

First, the deletion of whole variables is used. However, the disadvantage of this approach is the loss of some interesting types of information containing an excessive number of empty fields. For example, *post_money_valuation*. If most of the fields of a variable are empty, then line-by-line deletion would mean the loss of most of the values of other attributes as well, even though they themselves always have a value filled in on a particular line.

⁴ This number of values corresponds to the number of unique values with identical keys in all datasets that have been linked. Due to the nature of the data (for example, one row relates to investment series A and the other to C and is for the same company), the data are multiplied.

⁵ Python library

Row-by-row removal, therefore, followed only after all columns with a significant amount of the empty cells had been removed. On the other hand, what is not used for empty values are methods for filling in values, such as imputation with statistical values (mean, median, or mode) or interpolation, i.e., filling in based on known values before and after the missing points.

This is because the application of decision trees to data could change the characteristics of their individual branches, leading to less accurate decisions, and the application of these methods would disturb the natural structure of the data.

8.1.1. Removal of the Columns with Unnecessary Information

The other columns removed, apart from those with a preponderance of blank fields, were those that were too detailed and also had no bearing on the question under investigation. These are, for example, telephone numbers and unique identification keys, which only serve to link datasets together but say nothing about investment in firms, et cetera. They have no direct predictive value primarily because there is no relationship with the target variable, and their values are entity-specific and unrelated to patterns in the data. Removing these variables will thus ensure better interpretability and performance of the model.

At the same time, there are columns that, by default, contain only one single value from the start, so-called constant features, which are variables with zero variability, meaning they have the same value in all rows (*DropConstantFeatures* — 1.7.0, n.d.). This often happens in the data used here because a particular version of the Crunchbase approach has not been used, so only one value is set in a column by default. These are the variables *type_funding*, *type_investors*, and *type_organization*, which have been marked as constant features and removed.

8.2. Transformation of Values

Before applying the algorithms, it is necessary to transform some columns and aggregate the values into larger units. This is due to how the scikit-learn library (*Scikit-Learn: Machine Learning in Python - Scikit-Learn 1.5.2 Documentation*, n.d.), which relies on creating artificial variables, works. Before deploying decision trees on a dataset, there are always as many artificial variable columns created as unique values in each variable.

Thus, instead of making decisions on a few columns, the decision tree can decide on tens of thousands of columns based on a single variable.

However, too high dimensionality increases the risk of including random noise in the model, as the decision tree will try to find relationships even between artificial variables with a completely insignificant impact on the data, and the tree is more likely to find a distribution

that is statistically significant only by chance (*High-Dimensional Data*, 2024). Thus, the risk of overlearning or reduced ability to generalize on unseen data increases.

For variables where the values can be aggregated into a smaller number, it may be a good idea to do so and reduce the risks mentioned above. In this case, we will adjust for the categorical variables with the highest variability, namely the business category and geographic variables.

8.3. Category of Business

The business categories in the Crunchbase database are pretty extensive. Crunchbase records over 40 different groups of activities, and a company may list more than one of them at the same time. There are 26,060 in our dataset, and for an even more detailed division of business categories (in the *category_list* column), we record 91,019 unique entries.

The reason for these numbers is the way the database works. It recommends that companies list 3-5 categories at a time in their profile to encourage the search algorithm to display the company as often as possible. Thus, for example, a software company developing a product for gastronomy will be listed in the database in both categories that are somehow related to it, and the *category_groups_list* column will thus contain the value *software, gastronomy*. Other times, it will be *software, gastronomy, sustainability*, and so on. See Table 7 for examples of combinations in both variables displaying business categories.

Table 7-Method of recording business categories in the dataset

category_groups_list(examples)	category_list (examples)
Manufacturing, Science, and Engineering	Advanced Materials, Nanotechnology
Energy, Software, Sustainability	Energy, Energy Efficiency, Energy Management, Software
Apps, Financial Services, Information Technology, Other, Payments, Software	Apps, Blockchain, Cryptocurrency, Ethereum, Information Technology
Media and Entertainment, Video	Digital Entertainment, Media and Entertainment, TV, Video
Hardware, Manufacturing	Hardware, Manufacturing

Source: Based on data from Crunchbase

In order to reduce this excessive dimensionality of these two columns, it was necessary to map which categories and combinations occur most frequently in both columns. Since it would be too time-consuming to map all possible categories and then reduce them to the nth number of headings, I used text string search using Python functions.

Furthermore, *category_list* was removed since the variable is just a more detailed version of *category_groups_list*. The remaining variable, *category_group_list*, was analyzed based on the frequency of occurrence of each text string. As seen in Table 8, if any of the words in a text string matched a word in a particular cell, that occurrence was recorded and added to the total number of occurrences of that string.

Table 8-Searching the most common categories and sorting them by *category_groups_list*

Words Combination	Number
Lending and Investments Financial Services Payments	96 708
Hardware Technology Biotechnology	292 833
Software Artificial intelligence Machine learning Data and Analytics Information Technology	379 104
Medicine Health Healthcare	131 916
Education	32 893

Source: Based on data from Crunchbase

Recall that there were 712,616 rows in the original data. According to this simple analysis, the technology sector has the largest representation, with the information technology, software, or artificial intelligence subsector dominating. Conversely, the health or medical sector is numerically weaker and, along with any word identifying the technology sector, yields only 29,238 positive results.⁶

As a final solution, the values were replaced with two new ones. Those cells falling into the technology area are replaced by the value "Technology" and the others by the value "non-Technology."

8.3.1. Geographical Variables

There are four types of columns for geographical variables: *country_code*, *state_code*, *region*, and *city*, with different variability.

The country with the largest representation of startups (from the organization's dataset), investors (from the investor's dataset), and countries where the investment round took place (from the funding rounds dataset) is the United States of America. Overall, however, most countries in the world are represented in the original dataset. In order to reduce the number of artificial variables, an aggregation of values into two – "US" AND "Non-US" was used.

⁶ The reason for this analysis is the aforementioned way Crunchbase records business categories. Although a company may do business as a software company or a healthcare startup developing a program for medical purposes, they will be recorded in the data under a combination of different words, for example, just Healthcare and Software.

Other aggregated variables will be the *city* variables, which reach thousands of unique values. All values were classified into two categories: "core-startup-city" and "non-core-startup city". "Core-startup-city" replaced those cities that ranked in the top 20 in the respected Global Startup Ecosystem Report by Startup Genome (*Startup Genome*, n.d.)⁷. The second value then replaced the other cities in the variable.

On the other hand, the *state_code* and *region* variables have been removed. For *state_code*, the values were either the same as the *state_code* variable or a more detailed division of a larger state entity such as the United States, and for the *region* variables, it was a mixture of values for which there is no single unifying rule, and so for this inconsistency, the attribute is also removed. Geographic names included cities, states, and specific entities such as county names in the United Kingdom and so on.

Table 9 - Examples of values in removed variables region, here region_organization

region_organization	
New York	North Carolina
California	Ile-de-France
Illinois	England
Dublin	Georgia
Virginia	Massachusetts
Zurich	Colorado
Texas	Pennsylvania
Florida	Geneva
Arizona	Goa
Tel Aviv	Sachsen
Leeds	Alabama
Oxfordshire	Stockholms Lan
Washington	Ohio
Luxembourg	Quebec
British Columbia	Nevada
Berlin	Zuid-Holland
Maryland	Karnataka
Tokyo	Southern Finland

Source: Based on data from Crunchbase

⁷ The Silicon Valley region tops the list. However, the city variables include individual cities instead of this entire region. Thus, the core-startup-city includes those cities that belong to it according to visitsiliconvalley.org.

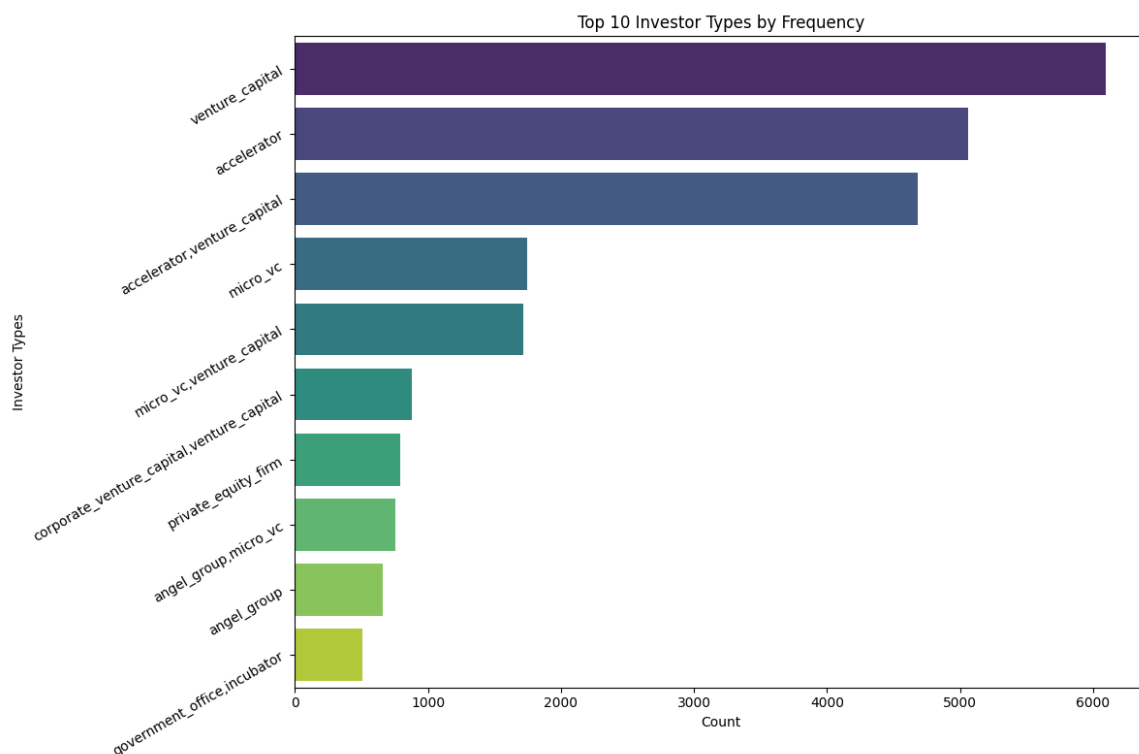
8.4. Types of Investors

There was a total of categories of investors in the original file, again due to how the investor type is recorded in the database. As can be seen in Figure 9, the investor type in the Crunchbase database actually consists of a sequence of words that characterizes the person or institution. For example, if it is a venture capital investor, its category will not only be *"venture_capital"* but *"corporate_venture_capital"* or *"accelerator, venture_capital"* and so on.

In order to reduce the number of artificial later in the application of the tree algorithms, it was necessary to sort all the original category names into three aggregated ones. The first aggregate category is called *'Incubators & Accelerators,'* the second *'Financial Investors,'* and the third *'Other.'* The keyword content replaces the original values. If a value corresponds to one of the three groups, it is assigned a new value covering that particular group of keywords.

The principle can be seen in Table 10.

Figure 9-Representation of investor types in the original dataset



Source: Based on data from Crunchbase

Table 10-Creation of summary category names for investor types

New Investor Types Category	Key Words
Incubators and Accelerators	• 'accelerator','incubator','university_program' • 'entrepreneurship_program','co_working_space' • 'startup_competition'
Financial Investors	• 'venture_capital','corporate_venture_capital' • 'private_equity_firm','hedge_fund','investment_bank' • 'pension_funds','fund_of_funds','micro_vc','venture_debt' • 'angel_group'
Other	• 'government_office'

Source: Based on data from Crunchbase

8.5. Logarithmic Transformation of Variables

Logarithmic transformation is a mathematical transformation that converts data values to their logarithms. Natural logarithms (base e) or logarithms based on 10 are most commonly used.

This transformation converts a wide range of values to a narrower, less extreme range (*Mathematical Biology Textbook: Nonlinear Data Transformations*, n.d.). Decision trees generally do not require interference with the data distribution and can thus learn from data sets that do not have a completely normal distribution. However, this is not entirely true. Indeed, possible outliers can significantly bias the algorithm's capabilities for several reasons: over-weighting for outliers, bias in the decision tree, and slow tree growth. The over-importance comes from the much greater influence of extreme values on the decision than values without extreme outliers. Trees may then focus excessively on these outliers, which may then be preferred by the model and accommodated by its decisions at the expense of generalization. This means overfitting (Echessa, 2024). Slow tree growth then occurs due to significant skewness in the data because it needs more splits to achieve the optimal result, and the tree will be larger and more complex, thus less efficient and more challenging to interpret (Kreinovich et al., 2024). How the variables in this original dataset performed in terms of skewness can be seen in Table 11.

Table 11 - Slope of variables in the default merged_df dataset before modification

	Skewness
rank_organization	3.59
num_funding_rounds_organization	2.18
total_funding_usd_organization	33.08
total_funding_organization	219.49
num_exits	17.77
rank_x	nan
rank_y	3.02
investment_count_investors	4.68
total_funding_usd_investors	8.04
total_funding_investors	159.83
rank_investors	-0.03
raised_amount_usd_funding	192.29
raised_amount_funding	637.4
post_money_valuation_usd	18.71
post_money_valuation	196.6
investor_count_funding	5.09

Source: Based on data from Crunchbase

The skewness of the variables refers to the degree of asymmetry of the distribution of values around the mean. It is important to examine it because high skewness can affect the performance of statistical models, especially those that assume a normal data distribution. A high skewness is considered when a variable takes on values greater than 1 or -1 (*The Complete Guide to Skewness and Kurtosis* | *Simplilearn*, n.d.). For machine learning, including decision trees, high skewness can be a problem because models may be prone to give more weight to majority values, which can impair generalization. The extreme values are unsurprisingly reached by financial variables that contain different amounts of money raised; after all, the dataset also contains so-called unicorns.⁸ Table 11 shows the skewness of the numeric variables present in merged_df. Logarithmic transformation will be applied to the following variables:

- *num_funding_rounds_organization*
- *total_funding_usd_organization*
- *num_exits*
- *investment_count_investors*
- *total_funding_usd_investors*
- *investor_count_funding*

⁸ a private company that has reached a market value of one billion US dollars (USD) or more without being listed on a stock exchange (i.e. before the IPO)

However, the rest of the variables will be removed from the data. This is for the following reasons:

Raised_amount_usd_funding and *raised_amount_funding* are redundant, as they are implicitly contained in the *total_funding* variables and, therefore, unnecessary to keep. The rank variables are redundant due to their specificity within the Crunchbase database; they are unstable, and there is no provision for how they are determined. Rank is a variable with little predictive power, as we do not know how exactly it is determined and cannot assess its relevance to regression tree models. Finally, *total_funding_organization* and *total_funding_investors* are redundant given that we have their equivalent between the variables but are additionally converted to the same currency: USD.

8.6. Modifying and Creating New Variables from Inappropriate Data Attributes

While some variables capture potentially valuable information, they cannot be used in their original form, and a new variable must be created from them. This brings us back to the issue of artificial variables. Several columns containing web links capture the presence or absence of firms and investors on social media. However, we transform these links into new variables and replace the original ones with these.

The importance of a company's presence on social media platforms has been declared in several academic studies, such as a large systematic study of 70 articles that focused on the impact of social media (Emmanuel et al., 2022; *Intangible Value and the City of London | Brand Finance*, 2023)

The study systematically examined the impact of social media (SM) on business performance, finding that SM use increases customer retention, relationship management, and financial results, especially for B2B companies and SMEs. It also helps network with stakeholders and attract investment. This positive effect was, therefore, also expected for businesses in Crunchbase data. New variables, namely:

- *Presence_on_social_networks_organizations*
- *Presence_on_social_networks_investors*

were created from the variables *facebook_url*, *linkedin_url*, and *twitter_url*. Each column with a filled value scored 1 point in a new column, giving a maximum of 3 points to a company or investor.

Table 12- Method of creating new variables from original columns in the dataset

Variable name	Method of evaluation	Values
Presence_on_social_networks_organizations	A point is awarded for a social network link in each column.	0-3
Presence_on_social_networks_investors	A point is awarded for a social network link in each column.	0-3

Source: Based on data from Crunchbase

9. Dataset Scope Limitation

The scope of the dataset is also limited due to the focus of the thesis, namely on firms referred to as startups.

First, it was necessary to limit the data to organizations in the role of a company. To do this, the variable *roles_organization* was used, which defines the roles held by the organizations, among which it is possible to find universities or companies acting as investors. Therefore, only those values containing strictly the word 'company' were retained, and all other values were excluded.

Table 13- Roles of the organizations in the original dataset

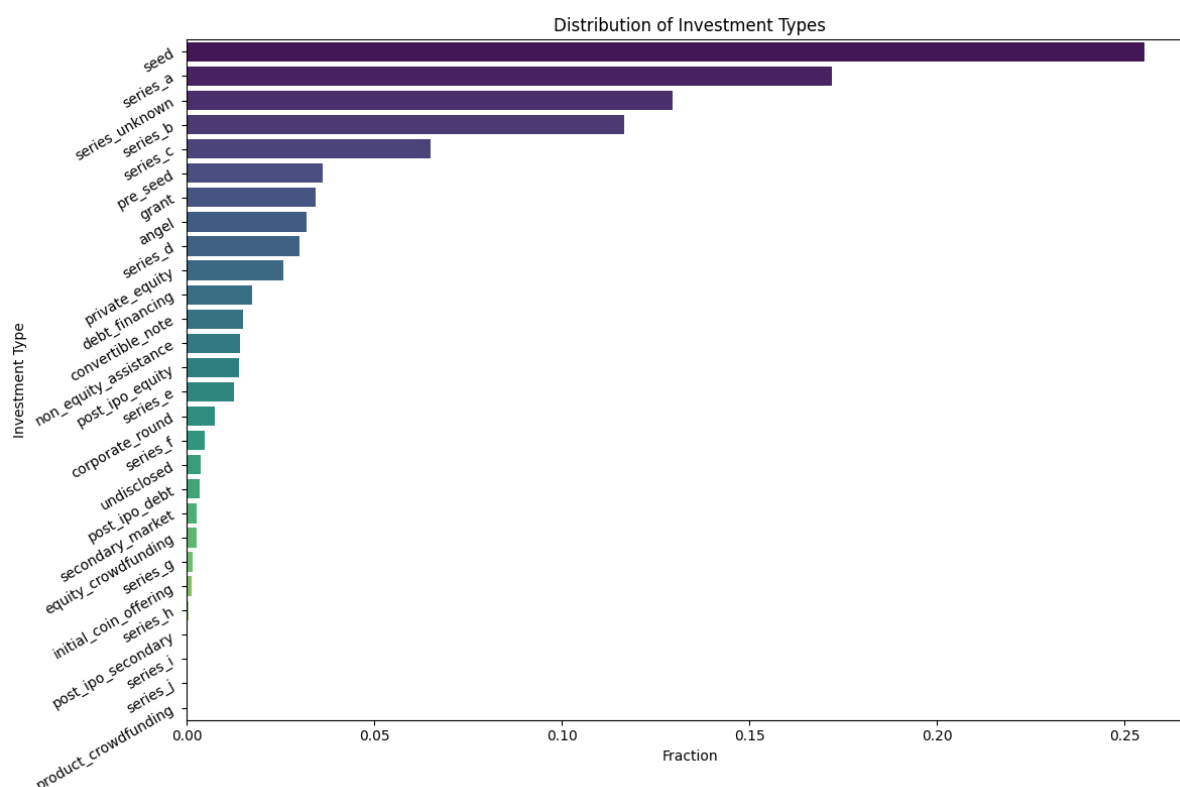
roles_organization	
company	investor, school
investor, company	school
company, investor	investor, company, school
company, school	school, company
investor	school, investor

Source: Crunchbase

The second limitation of the data is the type of capital financing the enterprise. Understandably, not all financing types fall into the early-stage investment category.

The largest share of the financing types is accounted for by so-called investment series and seed funding. Marquee Equity's website defines these as “a form of venture capital financing where a start-up company receives investments from venture capital firms or other investors in multiple rounds of funding or series”(What Is Series Funding and How Does It Work?, 2023). Here in the data, capital is limited to Series C.

Figure 10-Types of investment rounds



Source: Crunchbase

There are several reasons for this. First, Series C startups are still relatively young companies in a dynamic phase of growth and expansion. Second, later funding rounds are somewhat more complicated. Although a significant number of Series C firms exit funding, some firms continue afterward. However, these are generally more stable firms with a lower risk profile than in the early stages of the business, for example, looking to strengthen their positions and expand before the first IPO or have just not yet achieved the goals they set in the previous series.

The companies that thus undergo Series D and E are mainly already established businesses and meet the definition of a scale-up⁹, rather than a startup.

In addition to investment rounds, several other types of capital will be included in the type of capital retained in the dataset: angel investors and crowdfunding. Crowdfunding is an alternative to large investment rounds where firms raise money from individuals. It is mainly used by startups in the early stages of their business. Angel investors, according to Investopedia, "are early-stage investors in a startup. These investors provide seed money, usually in exchange for ownership equity in the company (Akhilesh, 2024) ." This type of capital is thus likely to support those companies that are being pursued.

⁹ Another stage of startup. Companies that have grown over the last three financial years at an annual rate above 20 % in terms of number of employees or turnover(CORPORATIVA, n.d.).

There are other types of capital in the data that could theoretically be associated with investing in startups. However, it is more likely to be a mix of companies in which they invest. These include private equity or initial coin offerings. Finally, rows that do not specify what type of investment it is, defined as "series unknown" in the Crunchbase database, are also excluded from the data.

9.1. Limiting Data to Specific Years Only

The necessary modification also consists of removing unwanted values for date variables. The first reason was the presence of historical data, which contained a total of 1777 firms with a founding date before 1900. Moreover, the decision trees trained later in the practical section do not want to learn on 19th-century data or older. The firms founded back then also reflect the financing methods of that time, which cannot be fully applied to current startups. Companies founded before 1900 operated in a completely different legal, regulatory, and financial environment. Modern startups face a very different environment, from access to venture capital to the impact of digital technology. Thus, discarding this pre-1900 data will ensure that decision tree algorithms do not learn from these irrelevant values.

The second data limitation will also be the year that can be considered a key point in the development of modern venture capital and when a specialized financial sector focused on financing startups or various technology companies began to develop and take shape. This year is considered here in this thesis to be 1960.

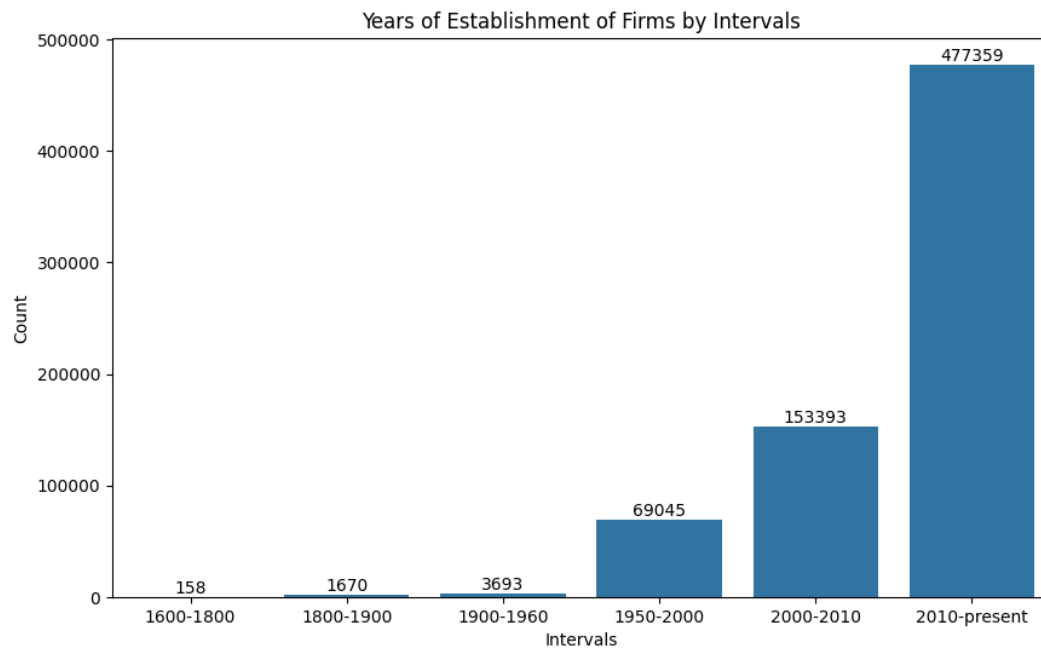
No particular year has been officially designated as ground zero for the current form of venture capital investment firms. The first real venture capital firm was the American Research and Development Corporation (ARDC)¹⁰¹¹ and its groundbreaking investment in Digital Equipment Corporation (DEC) in 1957, which many consider the beginning of modern VC. The development of venture capital broke out in full in the late 1950s and early 1960s, when the first venture capital limited partnership was formed - Draper, Gaither, and Anderson, among others, in 1958 (Nicholas, 2019, pp. 6-7).

It was in the 1960s that VCs began to focus on investing in start-up and expansion companies rather than investing in existing companies, and the emergence of now prestigious VCs such as Kleiner or Sequoia Capital (Gompers & Lerner, 2001).

¹⁰ ARDC's \$70,000 investment in DEC grew to \$355 million - 500 times ARDC's initial investment - when DEC held its initial public offering in 1968

¹¹ The company was founded in 1946 by Massachusetts Institute of Technology (MIT) President Karl Compton and General Georges F. Doriot (a professor at Harvard Business School), along with several local entrepreneurs

Figure 11- Years of the establishment of companies in the original dataset



Source: Crunchbase

In addition, the remaining years are grouped into individual decades. Dividing the time into decades reduces the number of unique values in the time column. Instead of single years, wider intervals are presented to the model, which can simplify learning and reduce the risk of overfitting.

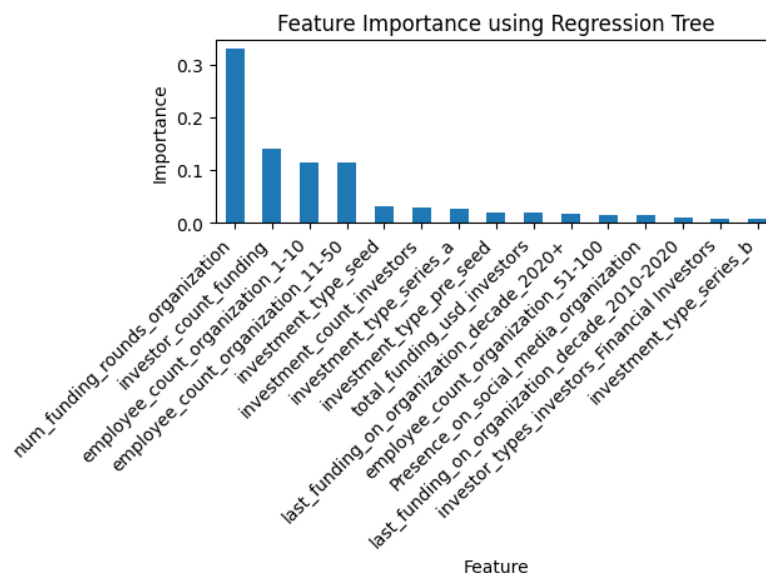
10. Regression Decision Trees - Analysis

10.1. Simple Decision Trees

10.1.1. A Tree without Hyperparameter Adjustment

The first group of models that will be used to identify key variables affecting the target variable *total_funding_usd_organization* are simple decision trees in the basic version and then in different modifications (different values of hyperparameters)¹². The first model is a decision tree with hyperparameter values preset in the scikit-learn library¹³. However, after fitting the algorithm to the training part of the dataset, the model achieves almost perfect accuracy, where R^2 at 0.994¹⁴ explains almost perfectly the variability of the target variable, and the model's error rate is minimal. In particular, the pivotal MSE metric computed at the nodes of the decision tree is only 0.03, which, knowing that the metric penalizes large errors due to squared amplification, means that the model would be nearly perfect. However, the model then deteriorated significantly on the test set. The coefficient of determination R^2 is at 0.67, and the MSE equals 1.72. It is also interesting to look at the RMSE to get the model's error rate in the original units of the explained variable,¹⁵ which jumped significantly from 0.18 to 1.34. Thus, the tree is significantly overfitted and, hence, not entirely appropriate for making decisions about which variables to focus on for startups.

Figure 12- Feature Importance of a simple decision tree without hyperparameter adjustment



Source: Crunchbase

¹² Every algorithm is present in five versions: basic one, three modified version and optimized model

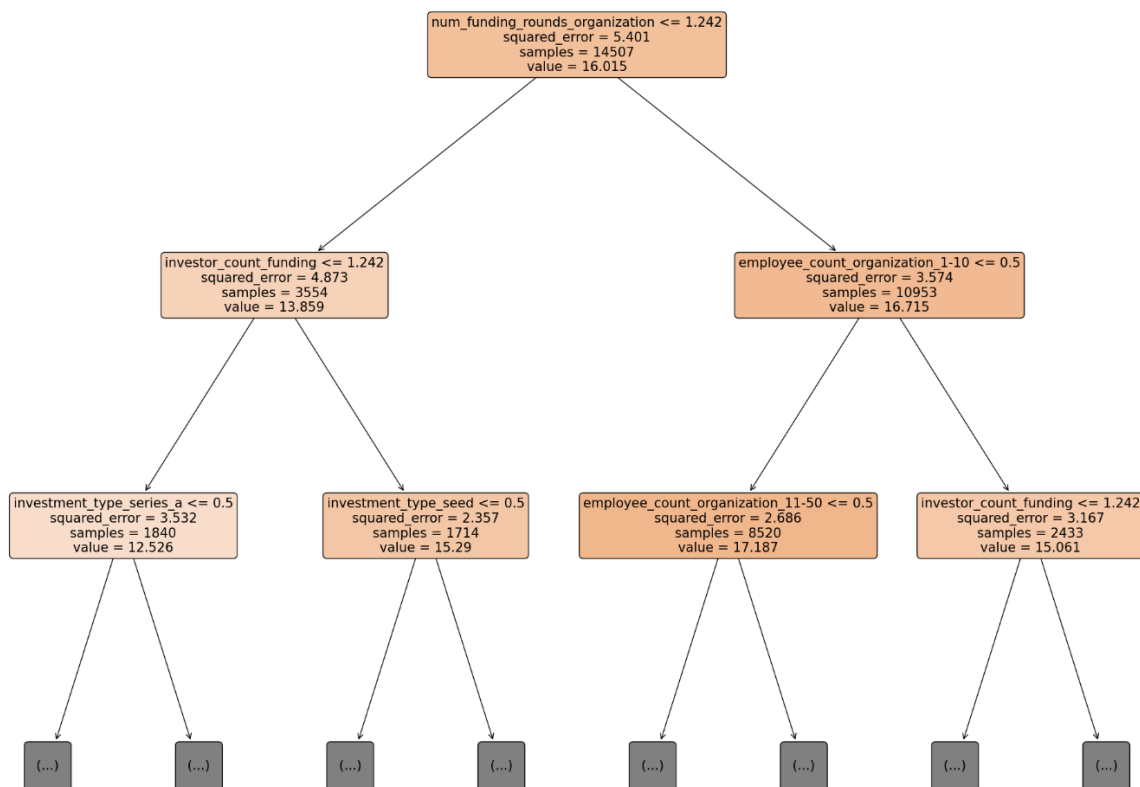
¹³ For more information see the appendix in the Basic settings of decision tree algorithms hyperparameters

¹⁴ If the value is close to its ceiling (1 for R^2). Or differs only in thousandth, the values are given in 3 decimal places

¹⁵ For all models in this paper, the explained variable is logarithmic

The tree has one dominant attribute, namely the number of investment rounds, with an importance of over 30%, which is not entirely surprising as we can expect to see *total_funding_usd_organization* growing along with *numb_funding_rounds_organization*. According to the significance of the flags, however, startups should focus on the number of investors in the investment round, the significance of which (*investor_count_funding*) exceeded 10%. Several artificial variables created from the default *employee_count* are also important to the model, but other variables play a minor role. Surprisingly, social media presence does not seem to be extremely important for startups, yet according to several sources, social media profiles are essential for a them (Gloor et al., 2018). Also, neither geographic variables nor business sectors play a major role in the model.

Figure 13- Simple regression decision tree model 1



Source: Based on data from Crunchbase

The first two layers of the tree indicate that the higher values are reached by the right branch of the tree, which uses the number of startup employees to divide after the root node. However, due to the depth of the tree, which has 36 layers, it is impossible to visualize the whole tree. However, using Python code, it is possible to determine how many regions and, therefore, how many different tree paths and values of the target variable *total_funding_usd* exist. This can be determined by the number of leaf nodes, of which this model has 12,602.

Thus, for a model of this unrestricted scope, the advantage of simple interpretation is clearly lost, as it would be very challenging to go through all paths and combinations, and a considerable amount of scope can be expected for a more significant number of the following models, for each of them individually. Therefore, I do not subject the regions and the relationships between their decision nodes to a deeper analysis.¹⁶

10.1.2. Simple Decision Trees with Hyperparameters

Whether the prediction ability in the decision tree has the potential to improve can be verified by changing the values of its hyperparameters. For example, if the tree is too deep, it may lose its ability to make reliable predictions on the test dataset due to noise adaptation in the data. However, insufficient depth for a change may cause the algorithm to be unable to capture more complex patterns in the data. Thus, the point is to find ideal values, and Table 14 shows the changed values of the simple regression tree hyperparameters for the three new models.

Table 14- Values of hyperparameters models 2 to 4

Attribute name	Model 2	Model 3	Model 4
max_depth	3	5	10
min_samples_leaf	1	5	4
min_samples_split	2	2	10

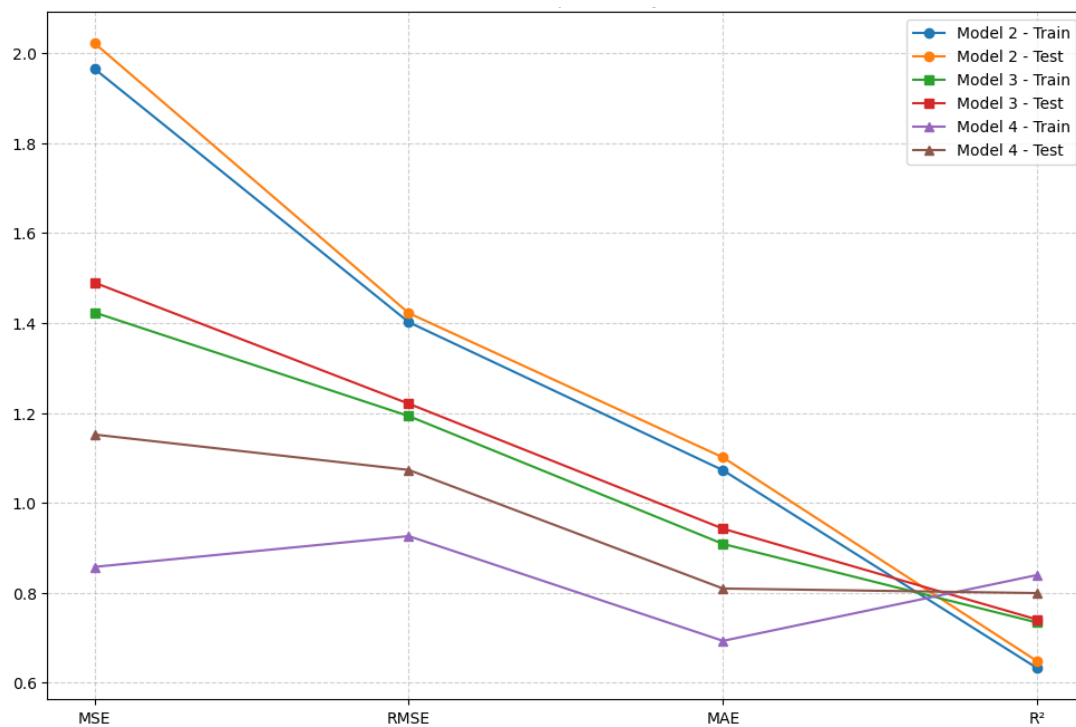
Source: Self-created

Figure 14 shows a comparison of the performance of the three models based on the performance metrics on the training and test sets. Which model is the best among them? Model 2 shows the worst performance on all 4 metrics with the lowest variability and, conversely, the highest errors. The lower value of R^2 , relative to the results of the other models, may indicate underfitting caused by an insufficient number of nodes to capture complex relationships in the data. High values of MSE, and hence RMSE, then penalize the model more for large errors. Model 3 has consistent results between the training and test sets and has a greater ability to explain the variability in the explained variable. The best explanation of R^2 is model 4, which also has the lowest error metrics. However, the two data sets have a more significant difference in performance. Since the goal is mainly to explain the dependencies in the data and it captures more relationships between variables, model 4 can be considered better.

Generalization, in this case, is not as crucial as the ability to identify relevant predictors, and the difference between `_train` and `_test` is not significant in absolute terms but only compared to the other two models.

¹⁶ The possibility of analyzing regions in large models is described in more detail in the Discussion section

Figure 14- Performance metrics for models 2 to 4



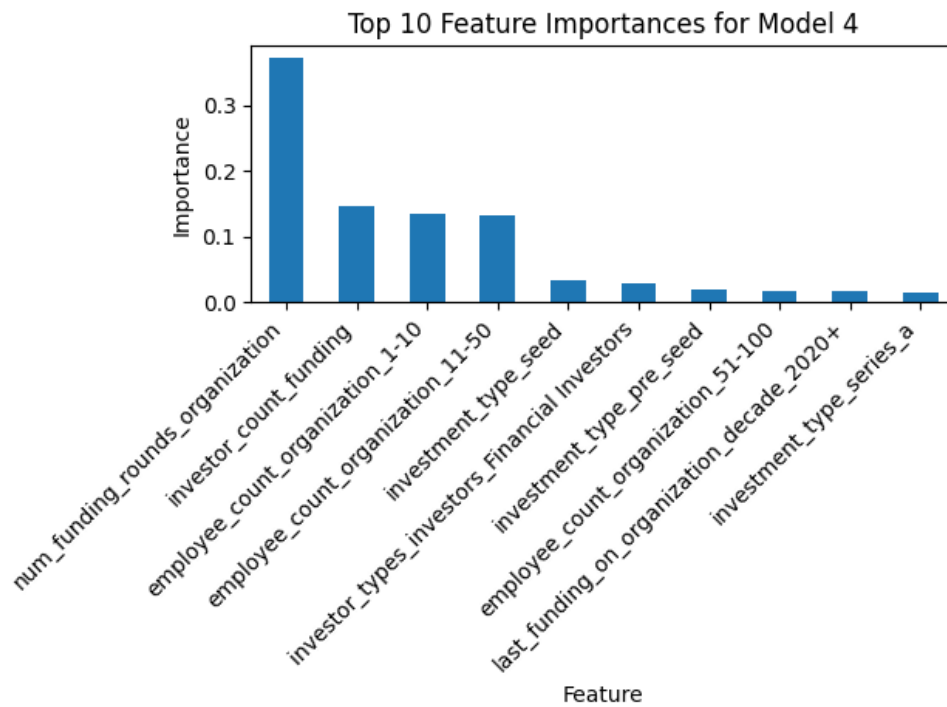
Source: Based on data from Crunchbase

Which variables contribute most to the resulting values of the target variable? Again, this corresponds to the importance of the feature and frequency of the occurrence of the variables in the nodes of the decision trees. In Model 4, the same features as in Model 2 are most important (worst performance among the three modified models). However, in contrast, the decade in which the firm last conducted a financing round also plays a role, with a preference for those conducted in 2020 and later. Geographical variables do not play a role in the other models either. In addition, in model 3¹⁷ versus 4, investment series b plays a role. In contrast, in model 2, the worst performer on performance metrics, the first 4 variables are the same, but the rest of the top 10 attributes have minimal to no importance, as shown in Figure 15¹⁸.

¹⁷ In Appendix

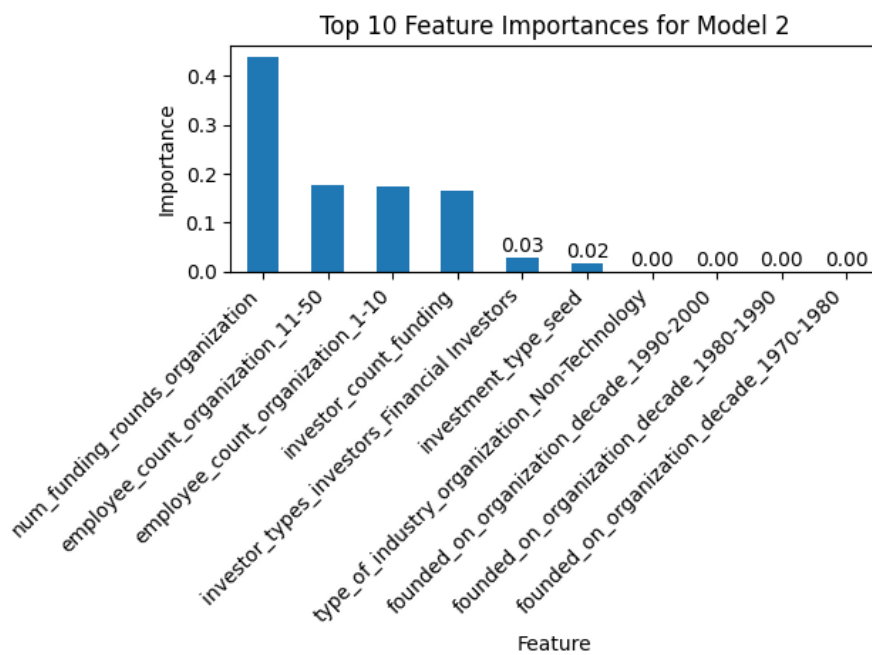
¹⁸ Detailed values in appendix

Figure 15- Feature Importance Model 4



Source: Based on data from Crunchbase

Figure 16- Feature Importance Model 2



Source: Based on data from Crunchbase

10.1.3. Grid Search and Optimal Model

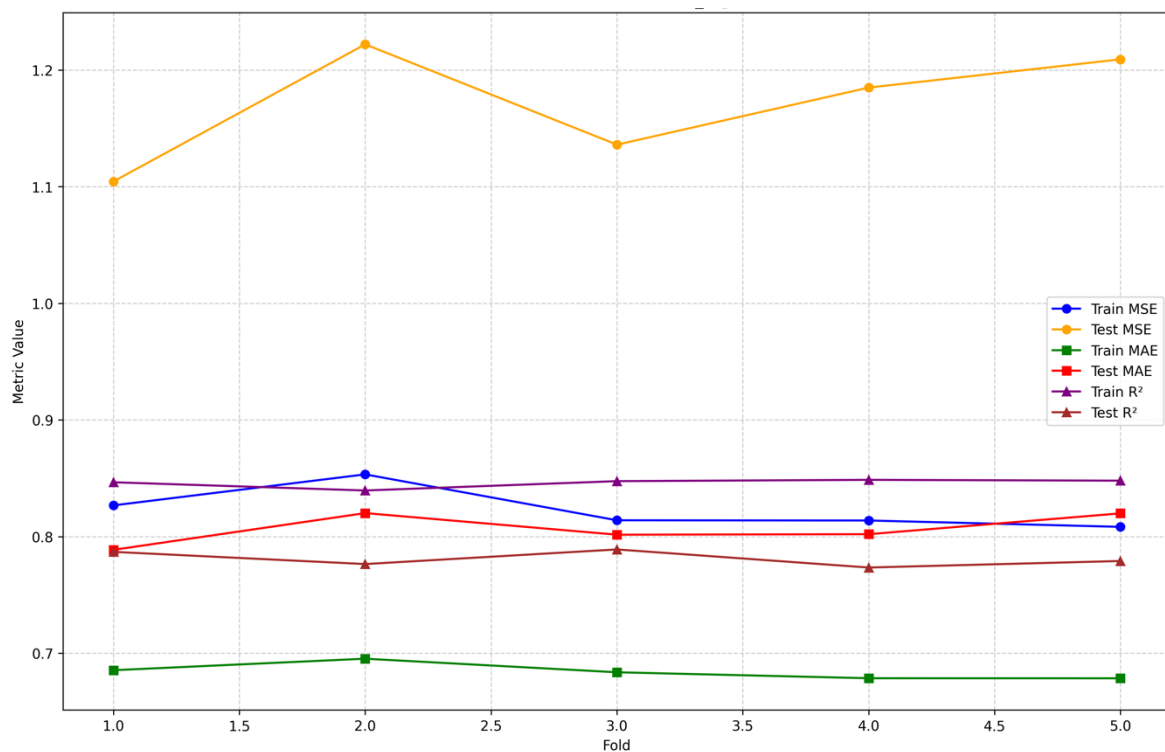
To find the most appropriate combination of hyperparameter values limiting the range of trees, while achieving the best possible results, the grid search method¹⁹ can be used.

For the hyperparameter values:

- max_depth=10
- min_samples_leaf=4 and
- min_samples_split=10

found in this way, the test metrics for the optimized model surprisingly perform even better than on the training set. For example, R^2 equals 0.67 on the training set, while 0.79 on the test set, and error metrics such as MSE are also reduced²⁰. However, this phenomenon may be due to the simple way of splitting the dataset at a particular point, which is not predetermined. Since the data is not completely uniformly distributed, the test set may then contain easier cases to predict, such as those behaving very similarly to the majority (little noise, few outliers), and thus cause this surprising situation.

Figure 17-Cross-Validation Results for Optimized Regression Tree



Source: Based on data from Crunchbase

¹⁹ The Grid Search table for a simple decision tree can be found in the appendix

²⁰ The values of the metrics can be found in the appendix

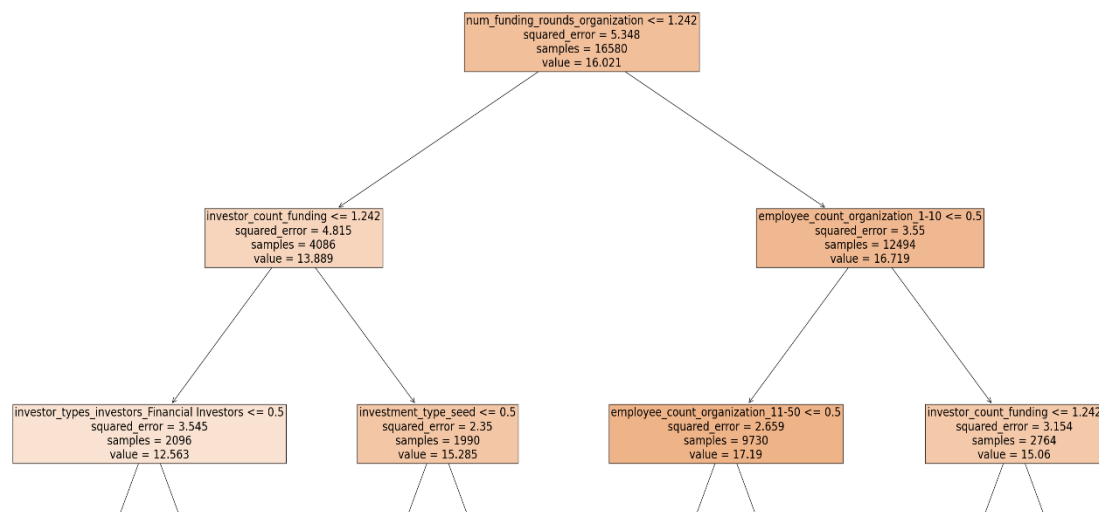
Table 15 -Resulting values of metrics of the optimized simple regression tree

Part of the data	MSE	MAE	RMSE	R ²
Train	1.79	0.94	1.34	0.67
Test	1.17	0.81	1.08	0.79

Source: Based on data from Crunchbase

The effect of the distribution of the set is also demonstrated by the cross-validation that tested the quality of the model, which shows that the distribution of the data plays a role, with metric values differing by two hundredths to one-tenth on average. Obviously, depending on the point at which the data is split, the results can vary greatly, and at each of the 5 folds, the MSE metric has a different value.

Figure 18-Visualization of the Optimized Regression Tree



Source: Based on data from Crunchbase

11. Ensemble Methods

11.1. Random Forest

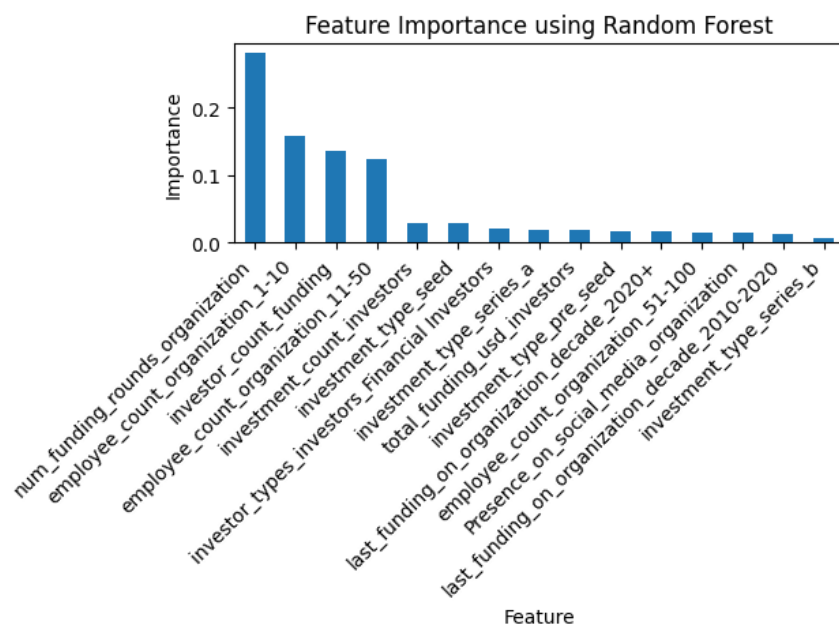
11.1.1. Default Model without Hyperparameter Adjustment

Simple decision trees do not achieve the best results, and there is a relatively significant difference in the results obtained from the training and test datasets. Therefore, from the point of view of accuracy, it is advisable to test the use of the ensemble methods first and then the random forest. Random forest should be applied to the dataset for efficiency on a larger number of variables; it should also better withstand noise.

The algorithm's performance without modifying the hyperparameters is analyzed first²¹. The variability shows a significant improvement over the baseline version of the simple decision tree, as R^2 even on the test set reaches 0.83, while on the training set the variability was 0.97. However, there are differences between the performance metrics on the two sets (train and test), indicating overfitting, where, for example, the MSE increases from 0.15 to 0.98. However, a value below one still indicates a slightly smaller number of large errors in the model, which is positive for the prediction of the level of capital a company chooses.

11.1.1.1. Feature Importance

Figure 19-Feature Importance Random Forest

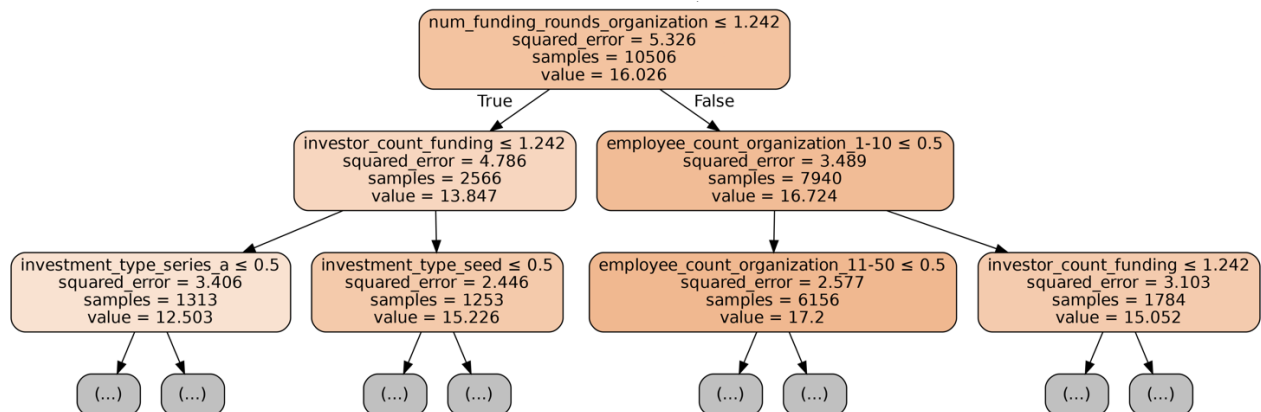


Source: Based on data from Crunchbase

²¹ A table with the basic values of the random forest investment hyperparameters can be found in the appendix

Even in the case of the random forest, the first four positions of feature importance have not changed significantly, and *num_funding_rounds_organization* has a decisive influence. However, among the variables with weaker influence, the variable *investor_types* appeared for the first time in a higher position in the top 10. Since it is impossible to visualize the entire random forest, a visualization of its trees is shown below in Figure 20.

Figure 20- Visualization of one of the random forest trees without hyperparameter adjustment



Source: Based on data from Crunchbase

Even the first layers of one of the trees show that more samples are associated with more funding rounds, although the *number_funding_rounds_organization* divides with a relatively high error and the tree apparently evaluated it as relevant to a particular subset of the data. Moreover, the error could be due to the fact that it is only the root node, where the error is usually higher. In addition, the relevance is based on the particular subset of data the tree is currently trained on, which is different each time, which is also why it is also true that comparison with other trees in the model would be difficult.

Given that this variable repeatedly appears significant in several other trees, its feature importance will also be high in other models. At the same time, these samples are split into their branch by *employee_count_organization*, and the left branch with fewer rounds uses the investment type.

11.1.2. Random Forest with Modified Hyperparameters

Even random forest predictions can be modified by setting the values of their hyperparameters. In a random forest, the combination of a large number of trees plays a crucial role, and each of them provides a different perspective on the data precisely because of the random selection of variables and samples. Thus, while a random forest has a natural resistance to overlearning due to aggregation of predictions, inappropriate settings can lead to an unnecessarily complex model or underutilization of data patterns. By optimizing these hyperparameters, one can try to find a better model in terms of prediction errors and the amount of variability explained by the model. The adjusted hyperparameters and their values are shown in Table 16.

Table 16- Hyperparameter values for modified random forest models

Parameter name	Model 6	Model 7	Model 8
n_estimator	100	500	600
max_features	sqrt	log2	log2
max_depth	10	30	50
min_samples_split	2	10	10
min_samples_leaf	1	4	10

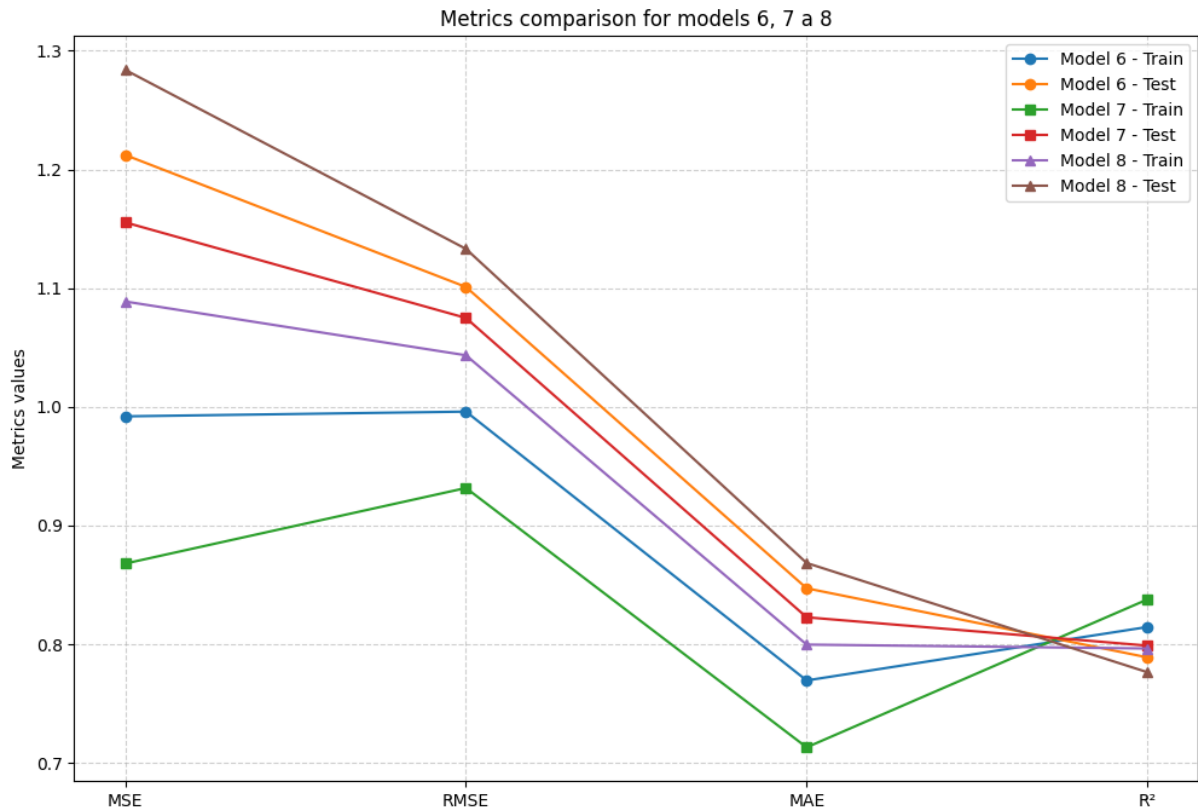
Source: Self-created

None of the models yielded substantial improvements in variability, and in some cases, as with Model 8, similar performance to some simple regression trees. This phenomenon is discussed more in the Discussion section. The `_test` and `_train` distances of the MSE metric values are interesting. Compared to the previous group of models with simple regression trees, the difference between training and test MSE is slightly higher (except for optimized regression tree model). The complex model shows higher overfitting, which is likely a combination of several reasons.

The random forest performs better in capturing non-linear relationships and interactions between variables. However, the test data contains more unseen combinations of variables, and the prediction will deteriorate. On the other hand, the error measurement models' performance metrics are lower, R^2 is relatively improved, and their values are more similar on both the training and test sets. When comparing only the individual random forests, the model without hyperparameter adjustment appears to be the best in terms of its ability to explain variability.

However, the same model has higher differences between *train* and *test* error metrics MSE, RMSE, and MAE. Thus, as an ideal combination of prediction accuracy and R^2 , adjusted models 6-8 are superior²².

Figure 21-Comparison of Metrics of modified random Forest Models



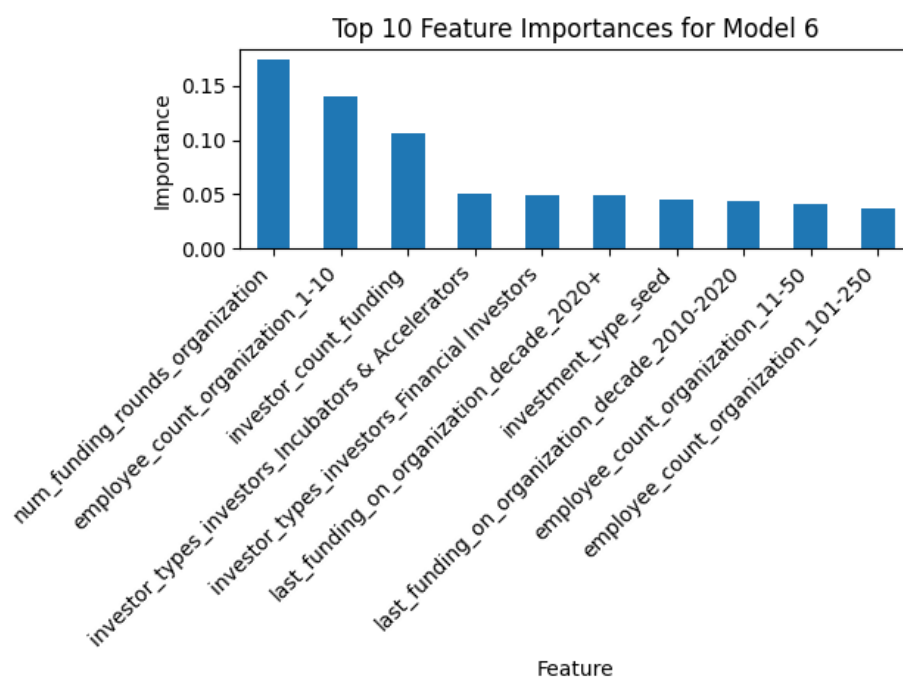
Source: Based on data from Crunchbase

11.1.2.1. Feature Importance of Modified Random Forest Models

In the case of the modified random forest models, the first 4 positions of feature importance did not change significantly in either case and again, *num_funding_rounds* has a decisive influence. No new variables appear among the top ten, i.e. no geographical variables or business categories. In particular, the significance of the first 4 variables in the 16-feature importance graph is significantly lower than the simple trees because the random forest more evenly distributed the significance among the variables due to the ensemble nature and random selections. It turns out that with the exception of the first three most significant variables, the remaining 7 variables are equally important for the amount of the explained variable *total_funding_usd_organization*.

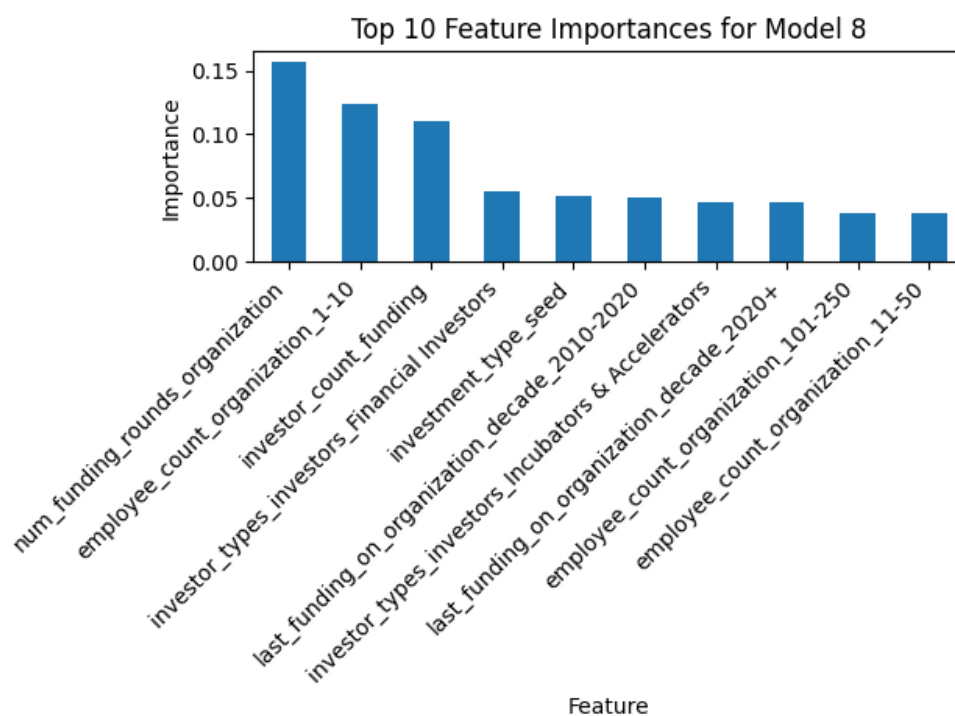
²² Detailed values in appendix

Figure 22-Feature Importance of Model 6 of the Random Forest



Source: Based on data from Crunchbase

Figure 23-Feature Importance of Models 8 of the Random Forest



Source: Based on data from Crunchbase

11.1.3. Optimized Random Forest

Even a random forest model, with the appropriate combination of hyperparameter values, can improve its ability to explain the variable's variability and reduce the metrics that calculate model errors. However, for the parameters `n_estimators=500`, `min_samples_split=2`, `min_samples_leaf=1`, `max_feature= sqrt`, and `max_depth= None`, the model did not yield any significant improvement or new features or change in their importance compared to the previous random forest models. The fact that the results did not change dramatically for the optimized model suggests that the relationship between the predictors and the target variable was already effectively captured in previous settings. It appears that the data itself may limit further performance improvements, and optimization would yield only minor changes in individual metrics and the balance between generalization ability and model fit.

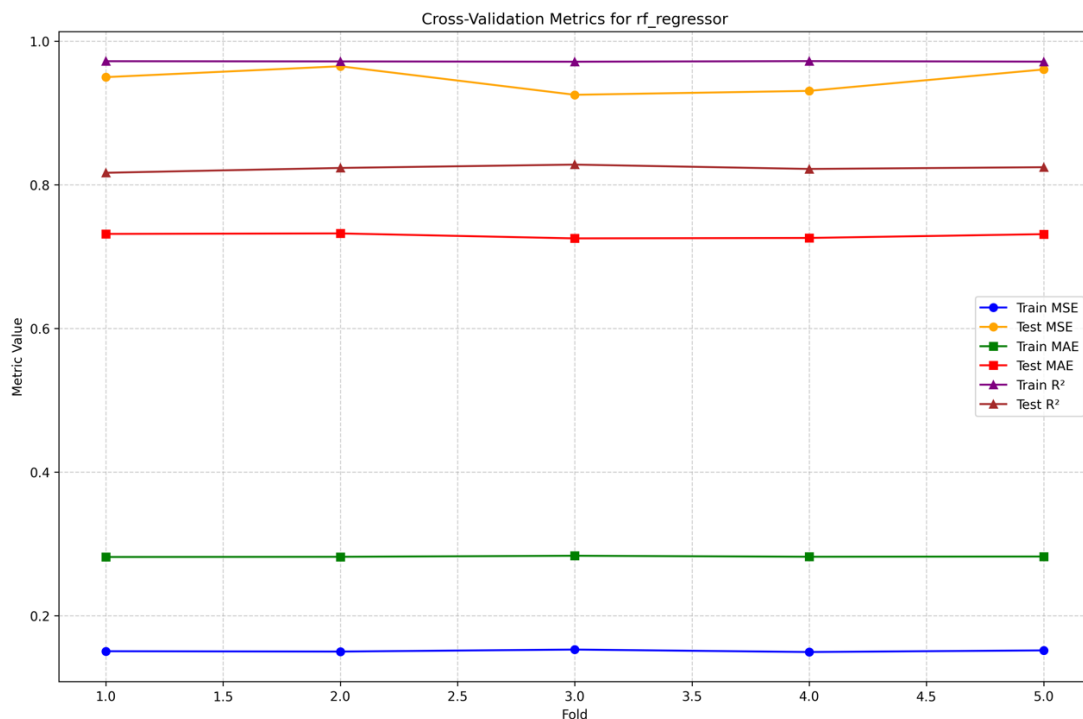
Table 17-Results of the optimized Random Forest model

Part of the data	R ²	MSE	RMSE	MAE
train	0.97	0.15	0.39	0.28
test	0.82	1.02	1.01	0.75

Source: Self-created

The fact that the model has reached its maximum is proved by cross-validation, which shows that there is virtually no shift in any of the metrics, no matter which part of the data the model is tested.

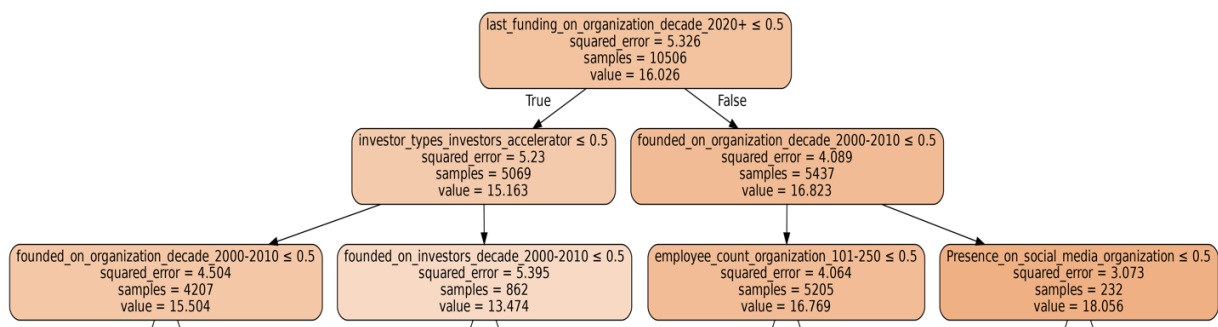
Figure 24-Cross-Validation of the Optimized Random Forest



Source: Based on data from Crunchbase

The visualization of a single random forest tree of the optimized model shows the importance of a deeper analysis of the ensemble method, where not only the most significant variable is the root node, but also, for example, *last_number_funding_rounds_decade* is the root node. Its presence may indicate that it contains more information relevant to prediction than it may seem at first glance

Figure 25-Visualization of one of the trees of the random forest



Source: Based on data from Crunchbase

11.2. Gradient Boosting

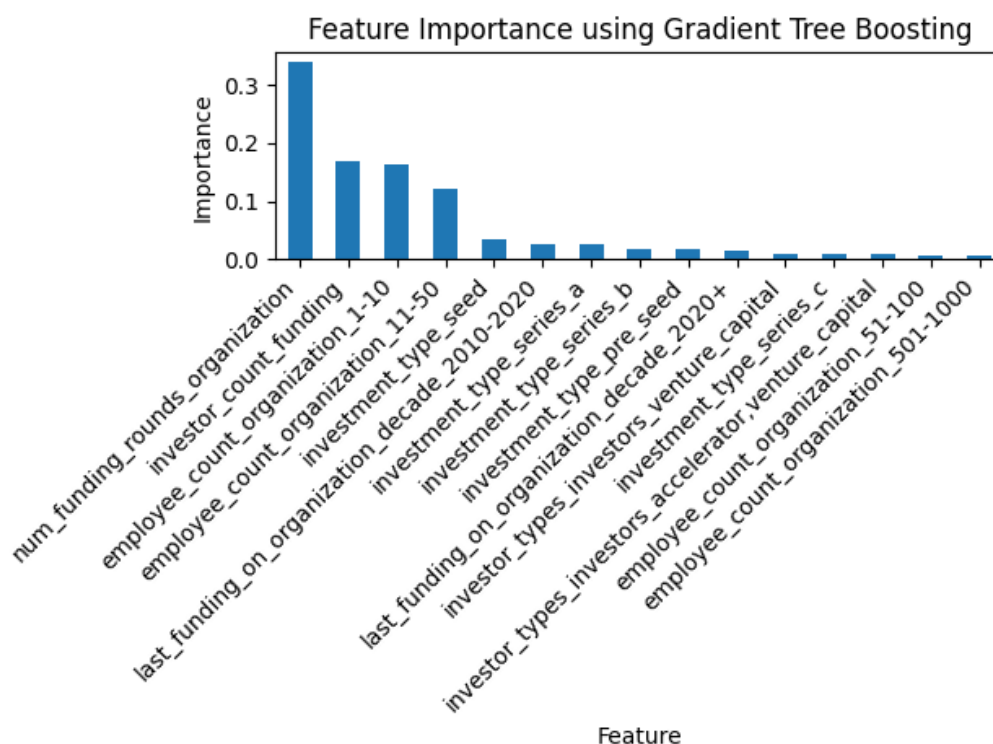
11.2.1. Default Model without Hyperparameter Adjustment

The random forest, in its various modifications, suggests some limitations, such as the amount of data and the ability of the algorithms to explain patterns in it with an acceptable level of overfitting. At the same time, no model has yet come up with a significant change in the order of the most important variables in the feature importance metric. However, it is possible that gradient boosting, where the new model learns from the previous model's errors instead of averaging the results of individual trees, may improve prediction accuracy somewhat in particular.

The model without hyperparameter adjustment²³ already shows more stability in the default version. The difference between the prediction errors is not very significant, with the MSE taking on 0.986 on the training data and 1.08 on the test data, the RMSE increasing slightly from 0.993 to 1.04, and the MAE of the model not increasing significantly either (from 0.76 to 0.80). The fact that MAE and RMSE are close to each other shows that outliers are not present to a large extent in the model. However, what is interesting is the very balanced value of R^2 , which hardly changes between the training and test sets, "dropping" from 0.816 to 0.812. The slight difference indicates the ability to capture key patterns in the data without noise adaptation.

²³ The model values are in the Appendix

Figure 26-Feature Importance Gradient Boosting Without Hyperparameter Adjustment



Source: Based on data from Crunchbase

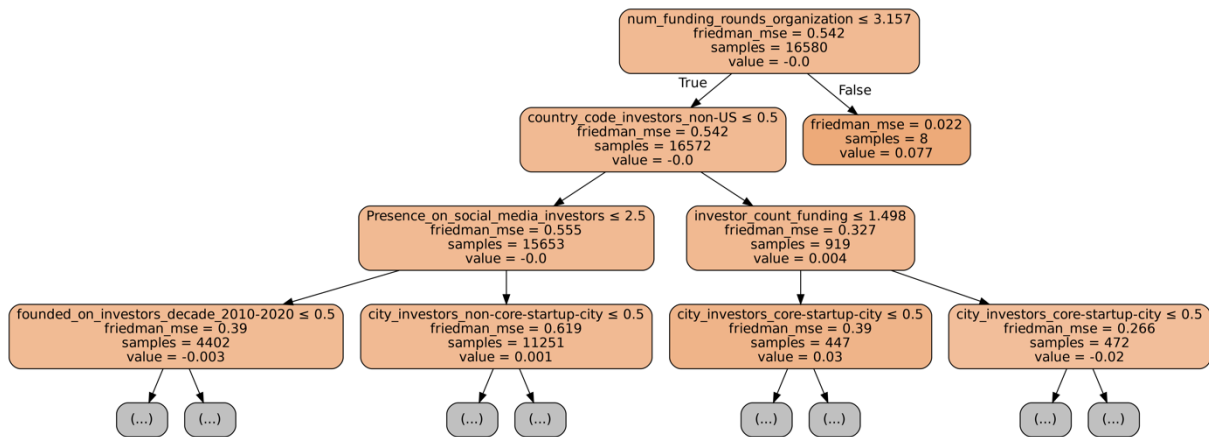
Cross-validation shows that the model would be stable even with a different dataset distribution, with MSE metrics ranging from 0.84 to 0.94.

In Figure 26, feature importance again shows the dominance of one variable over the others with a significance level exceeding 30%. However, the type of investor accelerator is now among the top 10 most important attributes, and the model thus more strongly emphasizes the influence of the investor and its type on the startup's success.

11.2.1.1. Tree Visualization Gradient Boosting

In contrast to previous models, it is impossible to see in the gradient boosting trees what values the explained variable takes on because the nodes show the magnitude of the correction of the explained variable. The model set the mean of the explained variable as 16.02. The root nodes and, for the purposes of this paper, especially the properties on which they are subdivided, play a crucial role, as they are always designed to minimize the error in each iteration.

Figure 27-Last tree of the Gradient Boosting model



Source: Based on data from Crunchbase

Of the 300 root nodes that select the property that best explains the error, 25 were split through *num_funding_rounds_organization*, and this attribute is the root node of the first tree as well. Meanwhile, in the beginning, the attribute selection is influenced by significant patterns in the data, while in later iterations, the model focuses on correcting minor errors. The visualization of this tree also shows geographic attributes, which have not yet appeared among the most important variables in the feature importance metric. There are 2084 of the 16,597 nodes in all trees of this model, and the trees make much more use of *city_* attributes (1384 nodes) compared to *country_code* variables (700 nodes). The meaning of the analysis of each node and the variables on which they are divided is more fully commented on in the Discussion.

11.2.2. Gradient Boosting with Adjusted Hyperparameters

Gradient boosting models contain other parameters compared to all the previous ones, which can be controlled to change the performance of the models and the selection of attributes based on which the decision-making process takes place. In the following modified models, the value of the learning rate, a hyperparameter that determines the magnitude of the error correction, is adjusted and is thus essential for balancing both the accuracy and stability of GBR.

Table 18-Parameters of modified gradient boosting models

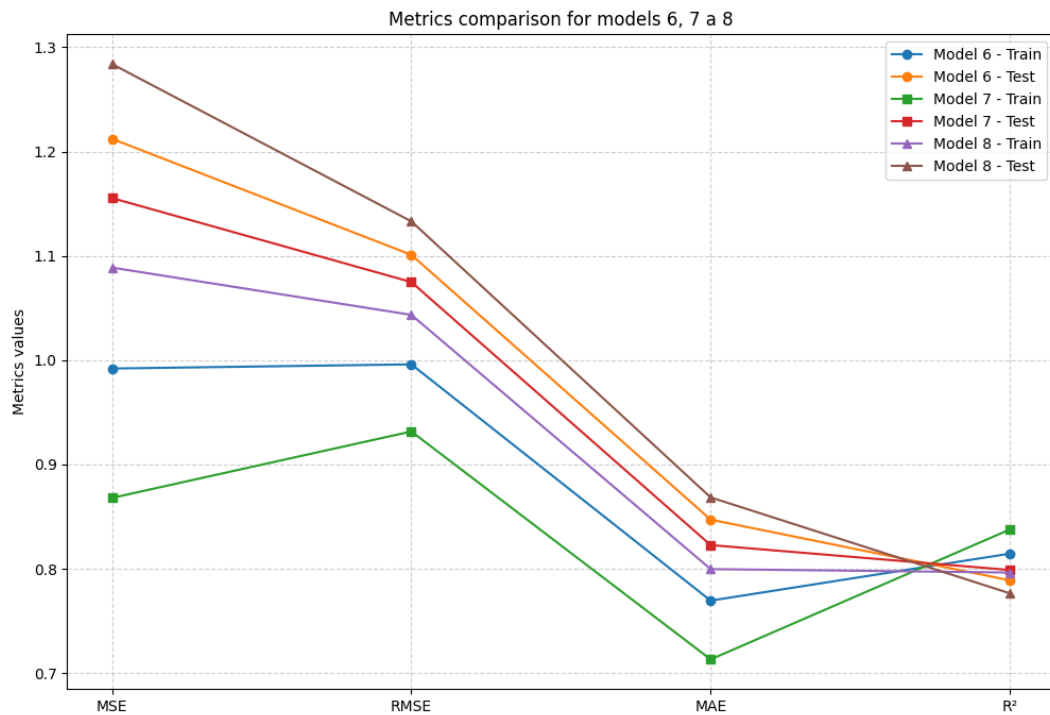
Parameter name	Model 10	Model 11	Model 12
n_estimator	100	500	600
max_features	log2	log2	log2
max_depth	3	7	10
min_samples_split	10	10	10
min_samples_leaf	4	4	4
learning_rate	0.01	0.2	0.04

Source: Based on data from Crunchbase

Figure 28 shows the effect of changing parameter values. Although the new values of the hyperparameters are intended to improve the model, in the case of the graph below it can be seen that this has not happened compared to the default model of the algorithm without modifying the values. On the contrary, in the case of model 9, for example, there has been a significant deterioration. Not only does it exhibit the worst error metrics ever of all the models, but it also has the worst explained dependent variable variability of 0.428. With such a low R^2 , it makes no sense to consider such a model. The bad values were probably achieved by inappropriate tree depth, minimum number of samples to split a node, etc. For example, the low depth of individual trees prevents complex relationships from being captured in the data. A low learning rate may also be a problem, which may only account for the small contribution of the tree to the overall model result. In contrast, with the best R^2 test result of 0.85 for model 12, there was only a slight improvement over the random forest²⁴.

²⁴ Detailed values in appendix

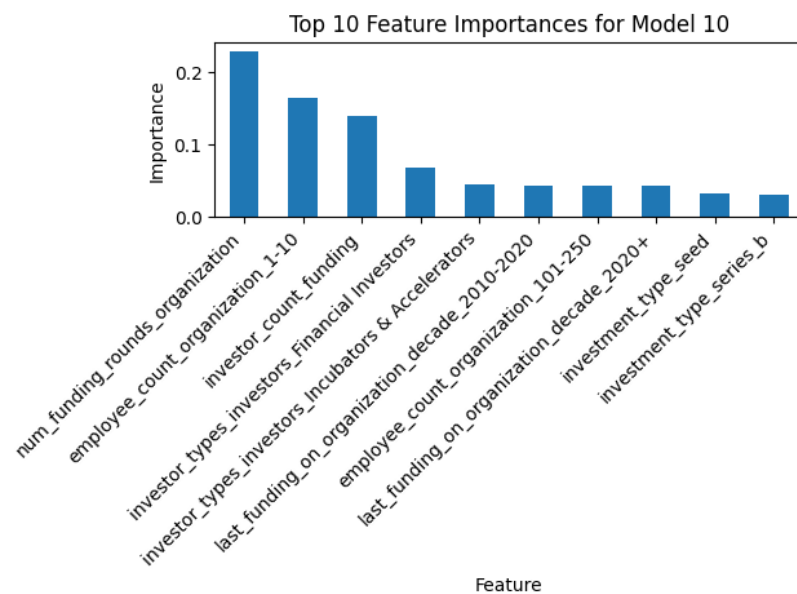
Figure 28-Metrics Comparison for Models 8-10



Source: Based on data from Crunchbase

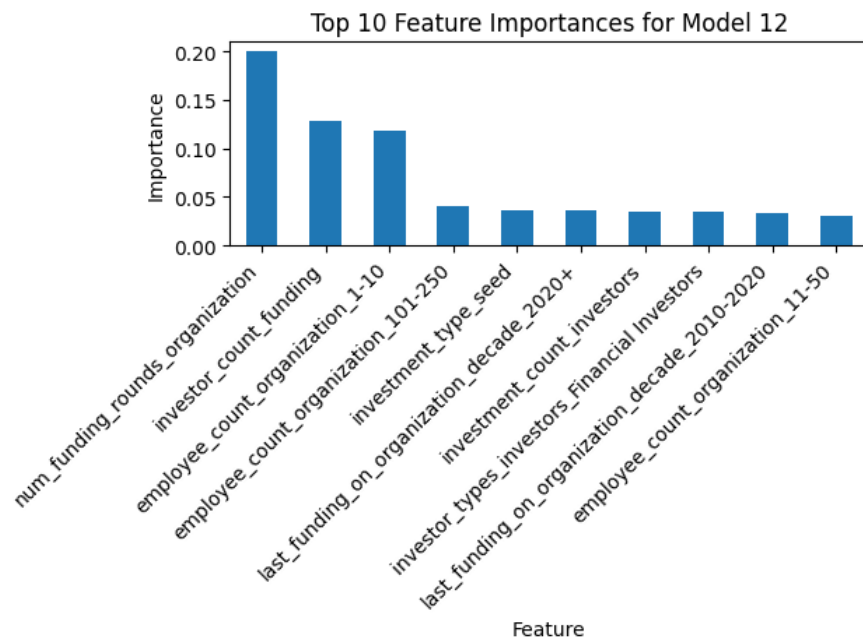
However, none of the three modified models have any new attributes in the top 10 variables that contribute most to model improvement. It might seem that this state of affairs is definitive. However, the Discussion section shows the opposite with the example of a different dataset modification, where this situation can lead to significant changes in the decision-making process results.

Figure 29-Feature Importance Model 10



Source: Based on data from Crunchbase

Figure 30-Feature Importance Model 12



Source: Based on data from Crunchbase

11.2.3. Optimized Gradient Boosting Model

In order to improve the model to the maximum possible level, specific hyperparameter values were again selected via the grid search method, namely $n_estimators=600$, $min_samples_split=2$, $min_samples_leaf=2$, $max_feature=sqrt$, $max_depth=10$ and $learning_rate=0.04$ but the model yielded almost no shift or new features or change in their importance compared to previous models. The R^2 value then even shows higher overfitting. Thus, we can assume that the model can no longer be substantially shifted under the circumstances.

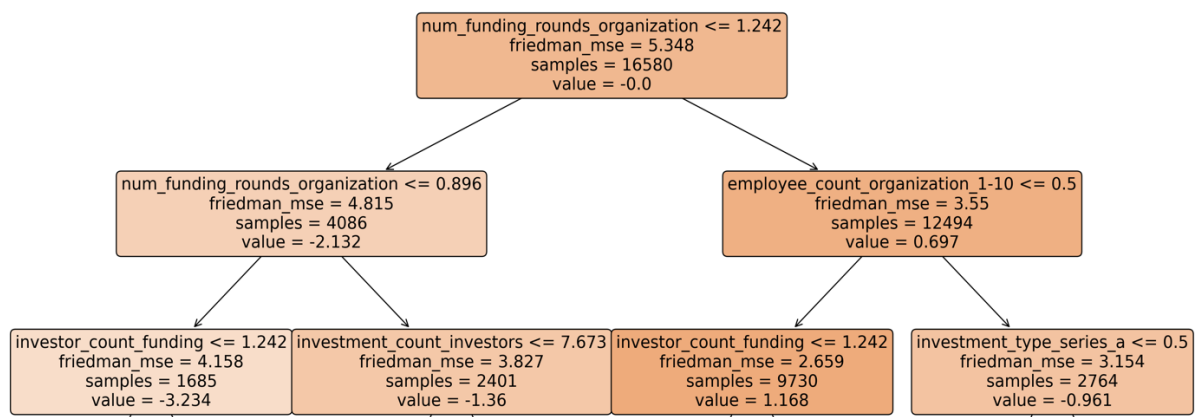
Table 19-Results of performance metrics of the optimized GBR model

Data set	MSE	RMSE	MAE	R ²
Train	0.398	0.631	0.442	0.925
Test	0.878	0.937	0.704	0.847

Source: Based on data from Crunchbase

The cross-validation of the model then shows that the value of the metrics of the optimized model is practically unchanged and very stable. It has achieved its best results for a given combination of hyperparameter values.

Figure 31-Visualization of the first tree of the optimized GBR model



Source: Based on data from Crunchbase

12. Discussion

Based on the results of all the models, it can be said that tree-based machine learning algorithms are used in predicting the factors that startups should focus on and using these machine learning analyses to reduce the risk of failure defined by the *total funding usd organization* variable.

At the same time, however, there are some drawbacks and situations to keep in mind that can have a significant impact on the performance of the algorithms. In general, tree algorithms are significantly affected by how the dataset is modified and are extremely sensitive to manipulation. For example, how empty values are deleted or how large categorical columns are reduced to those with only a few types of values also matters.

12.1. Summary of Model Performance and Interpretability

In this paper, it is evident that advanced algorithms combining the ability of relatively low overfitting, balanced feature importance, low model errors, and relatively high R^2 are most appropriate for the purpose of analysis. On the other hand, with an appropriate combination of hyperparameter values, even a simple regression tree was able to find a variation that can serve as a suitable complement to the analysis of factors affecting the future of startups.

Table 20- Summary table of algorithms performance(optimized) on train data

Algorithm	MSE	RMSE	MAE	R^2
Simple tree	1.79	1.34	0.94	0.67
Random forest	0.15	0.39	0.28	0.97
Gradient Boosting	0.25	0.50	0.32	0.95

Source: Based on data from Crunchbase

Table 21 -Summary table of algorithms performance (optimized) on test data

Algorithm	MSE	RMSE	MAE	R^2
Simple tree	1.17	1.08	0.81	0.79
Random forest	0.98	0.99	0.74	0.82
Gradient Boosting	0.87	0.93	0.69	0.84

Source: Based on data from Crunchbase

For all three types of algorithms, after applying the original unmodified version and testing various modifications, I finally applied the grid search method and thus let the algorithm protest all possible combinations of hyperparameters to find the optimal combination yielding the best results; those on the test data are summarized in Table 20.

In general, none of the tested algorithms substantially exceeded the 0.80 threshold for the coefficient of determination (R^2).

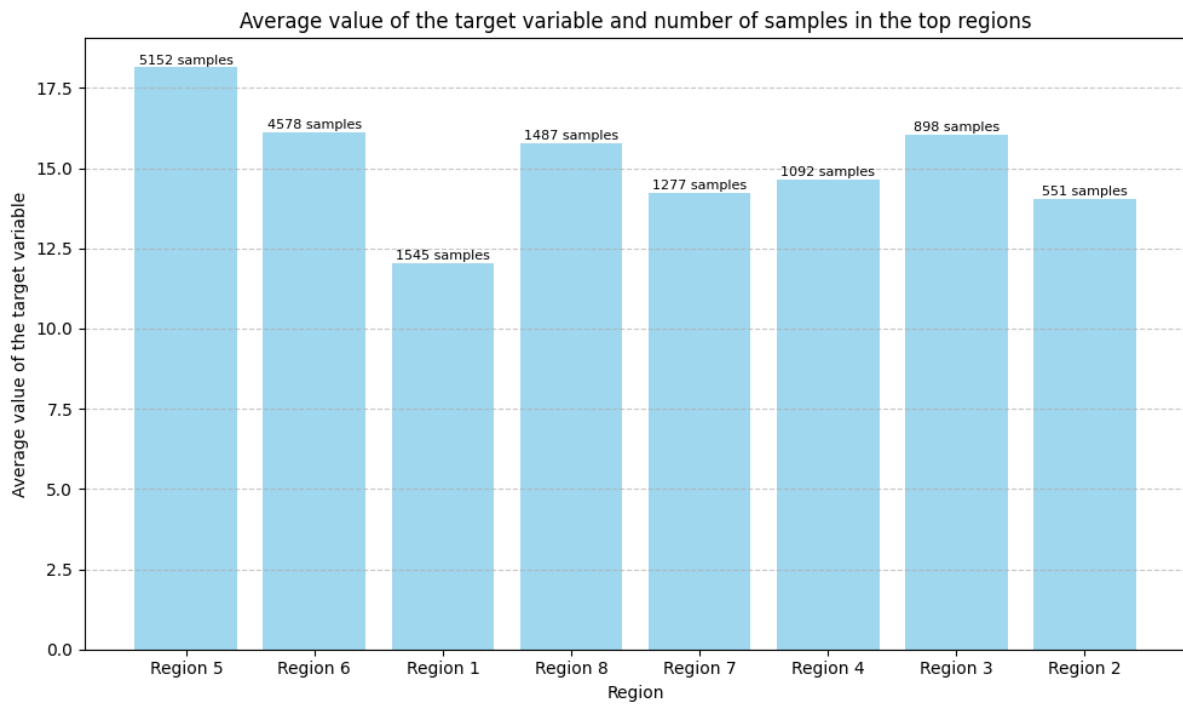
The difference, however, is that ensemble methods almost always hover around this threshold, while simple trees can fluctuate widely, with R^2 as low as 0.35 and as high as 0.60 on the test data. Thus, ensemble methods more stably eliminate any shortcomings, such as the random distribution of data at an inappropriate point.

At the same time, it should be noted that post-pruning, for example, was not applied due to limitations on the computational capacity side, which could potentially bring more interesting results to discuss even for simple trees.

Interpretability is also one of the key aspects of machine learning models, and decision trees are often described as easy to interpret due to their hierarchical structure that corresponds to the observed decision rules. However, this property is weakened in large-scale models, and this drawback is also shown in this paper. Indeed, for larger datasets, even for simple decision trees, the problem of high model complexity and the inability to visualize and subsequently analyze the models arises.

Even the simple decision trees in this work had over 12,000 leaf nodes, and the advantage of being able to analyze the combinations of variables and their values at the nodes that led to each leaf node is lost in such a large number. Those with fewer nodes then exhibit poorer performance for a change. This can be illustrated by contrasting two models, such as Model 2 (one of the modified models for which we randomly determined the values of the hyperparameters) and the optimized 'ideal' model.

Figure 32-Simple decision tree regions (model 2)



Source: Based on data from Crunchbase

It is possible to view the top regions in terms of the number of samples in a leaf node or perhaps the value of a target variable: in Figure 32, we can see all eight regions of Model 2, including the number of samples in each region. Similarly, the entire chain of combinations of properties and their values in a region that together lead to the target variable taking on a particular value can be displayed. However, this model only had a small number of regions due to the maximum depth of the tree being only 3, so the tree did not even have a chance to branch much, and the small depth corresponds to worse performance, with R^2 equal to 0.65 on the test set.

Figure 33-Top 5 Regions in Model 2

```
Region 1:
num_funding_rounds_organization <= 1.24 AND
investor_count_funding <= 1.24 AND
investor_types_investors_Financial Investors <= 0.50

Region 5:
num_funding_rounds_organization > 1.24 AND
employee_count_organization_1-10 <= 0.50 AND
employee_count_organization_11-50 <= 0.50

Region 6:
num_funding_rounds_organization > 1.24 AND
employee_count_organization_1-10 <= 0.50 AND
employee_count_organization_11-50 > 0.50

Region 7:
num_funding_rounds_organization > 1.24 AND
employee_count_organization_1-10 > 0.50 AND
investor_count_funding <= 1.24

Region 8:
num_funding_rounds_organization > 1.24 AND
employee_count_organization_1-10 > 0.50 AND
investor_count_funding > 1.24
```

Source: Based on data from Crunchbase

Thus, if we use a simple tree model with better performance, such as the regression tree model we just optimized, to visualize, we would get a model with an R^2 value of 0.79 on the test set but with 12,602 regions and a tree depth of 558. Thus, one must then set constraints on the parameters for selecting specific regions to analyze or accept the significant time and higher complexity of interpreting the tree.

Ensemble methods have the advantage of generally better performance, but it is impossible to visualize and interpret them in this way. This is certainly true for the random forest algorithm if we use a forest of hundreds or more decision trees, and their performance is averaged.

For the gradient boosting algorithm, the analysis through features essentially loses its meaning since each tree works on the principle of improving the previous model's performance, and analyzing the trees separately thus loses its meaning. Moreover, even for random forest models that average the performance of each tree, we are talking about 543,399 leaf nodes for model 7 alone, for example.

We can also count how many nodes of decision trees or ensemble methods were split based on a particular variable of interest or if we thought it was important.

Focusing specifically on the geographic variables examined in the context of visualizing one of the trees of the GBR algorithm and suggesting that they were not entirely insignificant, as they still accounted for over 10 % of the nodes of all trees.

This is consistent with the findings of various statistics tracking the success of startups with a detailed focus on the cities. StartupGenome, but other statistics also usually do not focus on entire countries, for example, the Netherlands, but specify Amsterdam. In addition to investors, talent, geographic location advantages, and so on, smaller entities may also differ in terms of tax burden or regulation.²⁵

However, this analysis would find more application in simple trees where it can be combined with the number of nodes in which it is found or which have been created by splitting nodes containing a particular variable and where their meaning is unambiguous. This is because, in the ensemble methods, it can be challenging to determine how much a particular node or region has an impact on the model result, either in terms of averaging in the random forest or the position of the tree in the tree chain in gradient boosting. Even in terms of identifying the causes of the value of the target variable, a simple decision tree is preferable, where the average value of the target variable in each region can be calculated and, together with an analysis of the attributes in each region, it can then be decided which features to focus on.

Table 22 - The value of the target variable in the regions of model 1²⁶

	region	n_samples	mean_value	impurity
4	Region 5	5152	18.14	1.61
5	Region 6	4578	16.12	1.67
0	Region 1	1545	12.04	2.15
7	Region 8	1487	15.77	2.43
6	Region 7	1277	14.23	2.71
3	Region 4	1092	14.65	1.45
2	Region 3	898	16.05	2.37
1	Region 2	551	14.04	4.52

Source: Based on data from Crunchbase

²⁵ Even if the dataset does not directly contain this type of information, it may be implicitly contained in the geographic variables.

²⁶ Impurity means the degree of variability of the target variable in each region of the tree

12.2. Scope and Data Aggregation

In this paper, the selection of features for models is limited to a narrower set of information, which limits the ability to find new key and perhaps surprising features for startups to focus on. Thus, the largely repetitive attributes in place of the most important characteristics could be supplemented by others if others were available in the data - such as gender, education levels of founders or other team members, more detailed financial information, data on patents acquired by firms, and so on. The importance of data diversity for ML models is described in their paper by Gong et al.(2019) as a sufficient range of discriminative information, thus improving generalization to unknown data.

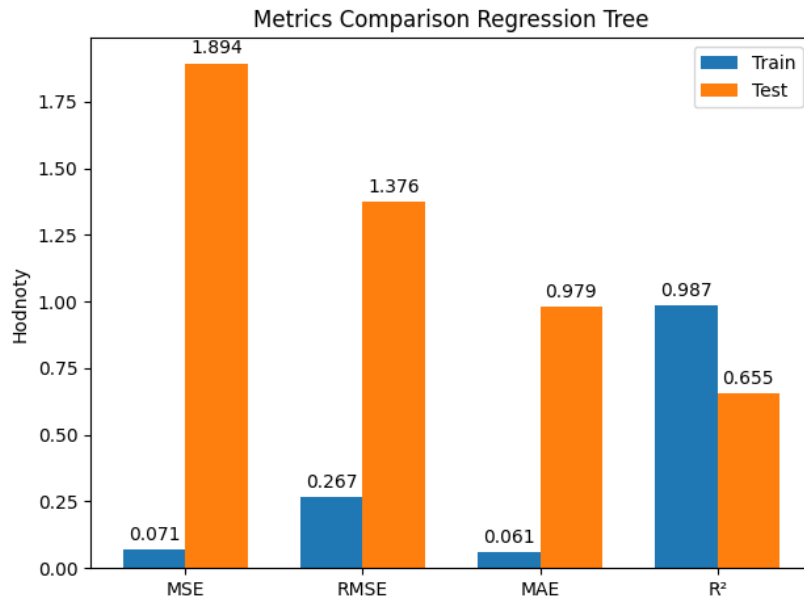
Feature engineering also involves the transformation and aggregation of existing information in the dataset, which has an absolutely critical impact on the performance and relevance of the model. The data modification in this thesis involved the aggregation of data in geographic attributes, time data into decades, or lines of business into two categories: technology and non-technology. However, with comparable models and the same hyperparameters used, the tree algorithms would have achieved very different, and often substantially worse, results.

Figure 34 shows the difference in error metrics and R^2 on data where only the business category attribute (the name of the variable) and the geographic variables were aggregated. The metrics have changed slightly, but most importantly, a new variable:

- *type_of_industry_organization_Non-Technology*
- *Presence_on_social_media_investors*

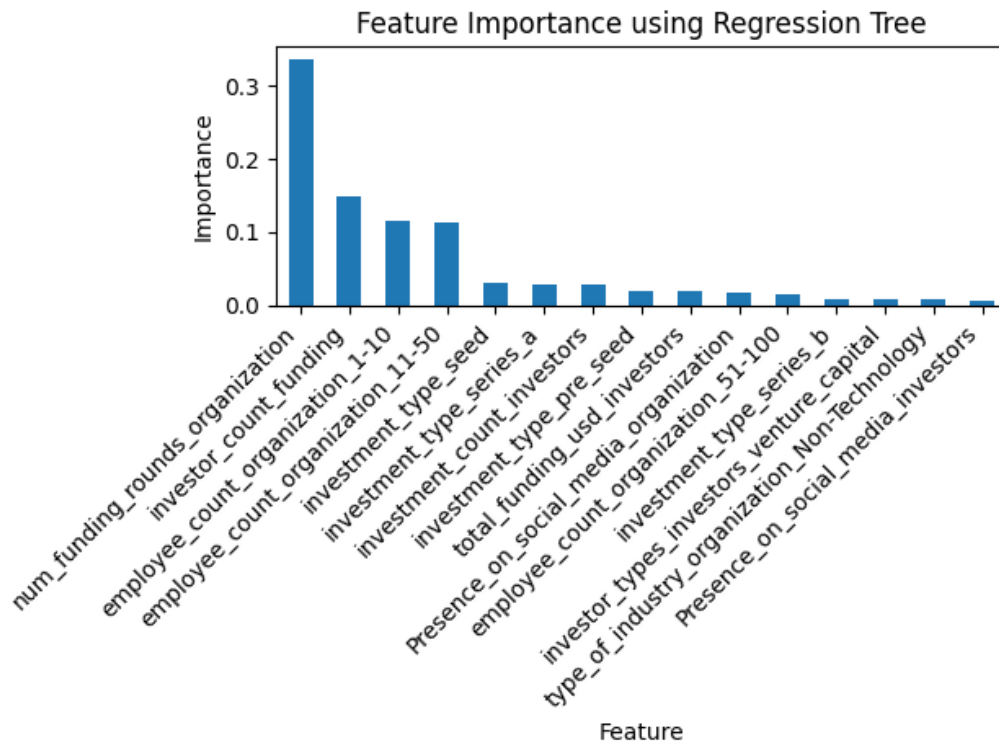
has been added to the same model among the top ten characteristics in feature importance. Similarly, the context of the repeatedly rejected hypothesis of the importance of geographical variables could also be considered. While these are never among the 10 most important for models, they are involved in decision-making processes in the most accurate ones in 13 to about 15% of nodes. Their importance is not explicit, but they may be implicitly linked to other key attributes, such as total capital, which may be more available if the firm is located in specific regions/countries. Thus, their presence may contribute to a better capture of relationships.

Figure 34-Metrics Comparison for Regression Tree after not Removing Outliers



Source: Based on data from Crunchbase

Figure 35-Feature Importance for Regression Tree without Aggregation



Source: Based on data from Crunchbase

12.3. From Research to Real-Life Impact

There is not much information available on the impact of machine learning on a startup's decision-making process, deciding which factors to focus on early in its existence. However, when it comes to focusing on the key factors for the business's success, the opposite side provides quite a lot of information, i.e., the investor who intends to value their money by investing in the startup - venture capital.

The involvement of artificial intelligence is not the music of the future. Although getting exact figures from investment companies is impossible, there are already some interesting observations. For example, Henrik Landgren, one of the partners at EQT Ventures, mentions the use of AI in an interview with the Wall Street Journal (Marco, 2023). The company uses an AI platform called Motherbrain, which the Stockholm-based firm developed in-house and is proprietary software²⁷.

The model combines features of language models and models that evaluate numerical data. According to Landgren, they had already made four investments at that time that they probably would not have made without the involvement of AI. The platform includes information on initial funding and financial information, as well as the history of team members or traffic on their websites. It uses a scale to assess the investment prospects of the projects in question, with one of the four projects already mentioned having been successfully divested at the time.²⁸

Correlation Ventures, which has set up a platform that analyzes data from more than 100,000 investment rounds and also sets up decision-making mechanisms via a scoring model, thinks similarly about AI (*Gartner Says Tech Investors Will Prioritize Data Science and Artificial Intelligence Above "Gut Feel" for Investment Decisions By 2025*, 2021).²⁹ Overall, companies are also using tools to determine business values or to set strategy, from long-term to operational plans.

²⁷ Proprietary software that is subject to copyright and contains restrictions on use, distribution and modification imposed by its publisher, vendor or developer. This type of software remains in the possession of its owner/developer and may be used by end-users or organizations under predefined conditions. Other names closed source or commercial software.

²⁸ Peakon, sold to Workday Inc. In early 2021

²⁹ 2021

In-house developed solutions are challenging to explore as we usually depend on information the organization provides without the opportunity to test and evaluate the tool.

Even for startups themselves, there are now a number of tools they can incorporate into their decision-making processes, whether it is recruitment, campaign development, or predictive analytics (Panel®, n.d.). Examples include Albert, Upmetrics, and Salesforce Einstein (*Top 20+ AI Tools For Startups*, n.d.).

13. Conclusion

The thesis, entitled *Application of the Tree Algorithms to Analyze Factors Affecting Startup Success and Assessing their Accuracy*, examines the applicability of tree algorithms for businesses referred to as startups at the beginning of their entrepreneurial journey and whether these methods are relevant tools that can be used to identify key factors for the success of the business and how accurate these algorithms are, and whether startups can therefore use these methods in their decision-making process. While the results of the analytical section show that ensemble methods demonstrate quite a stability compared to individual trees, where a single decision tree exhibits greater performance variability, they are very complicated in terms of interpretation. Additionally, stability is observed in ranking the most important variables across all types of algorithms. Nevertheless, that does not mean that the application of such algorithms can be recommended for use without completely without potential risks.

As can be seen especially in the Discussion section, just by choosing a different way of adjusting the data, the most important attributes may include other attributes than the ones used in this thesis, and therefore, geographical ones when tested repeatedly and deeply. Moreover, models can start making significant errors. At the same time, the performance of the models has reached certain limitations under the circumstances.

In particular, the more powerful random forest and gradient boosting (GBR) models achieve a coefficient of determination of over 80% but have a performance ceiling around this value. Again, as shown repeatedly in both the Data, Analysis, and Discussion sections, the results of tree-based algorithms highly depend on how the dataset is modified and its extent, the chosen values of the hyperparameters, or the chosen algorithms and programs applied to the data. For example, leaving or deleting empty values fundamentally changes the predictive value of the models. The consistency in the latter of the most important values may also show a potential for adding another variable, such as the gender of the founder or diverse financial indicators, which could lead to different model performance. Finally, tree-based algorithms cannot always work efficiently with significantly outlying values.

Thus, although it is not possible to make a clear recommendation on the use or non-use of tree algorithms for analyzing startup success factors based on one external analysis, the models can serve as a valuable complement to identify new and interesting patterns in the data that companies can then focus on.

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18. Appendix A

18.1. Meaning of All Columns in the Original Dataset Before Modifications

Column Name	Description
alias1	Other known names/alternate/previous names of the company
alias2	Other known names/alternate/previous names of the company
alias3	Other known names/alternate/previous names of the company
category_groups_list	List of category groups tagged to the company
category_list	List of category groups tagged to the company - more detailed
cb_investment	Crunchbase profile URL
cb_url_funding	Crunchbase link of the organization associated with the funding round
cb_url_investors	The Crunchbase link of the investor
cb_url_organization	Crunchbase link of the organization
city_funding	City associated with the funding round
city_investors	The city investor is located in
city_organization	The city organization is located in
closed_on_investors	Date of Investment firm's closing
closed_on_organization	The date that the company closed
country_code_funding	Country code associated with the funding round
country_code_investors	Country code of investor's HQ
country_code_organization	The country that the company is located in
created_at_funding	A time funding round was created.
created_at_investment	Date when information about the company was added on Crunchbase
created_at_investors	Time investor was created - investors can be people, companies, or investment firms.
created_at_organization	Date when the company was added to Crunchbase
domain_investors	Website domain of an investor
domain_organization	Website domain of an organization
email	General contact info for the organization
employee_count	Total number of employees
facebook_url_investors	Facebook URL of Investor
facebook_url_organization	Link to the organization's Facebook
founded_on_investors	Date Investor was founded.
founded_on_organization	Date company was founded.
funding_round_name	Name of the funding round
funding_round_uuid	ID used to identify funding round
homepage_url	Fully qualified URL of an organization
investment_count	Number of investments
investment_type	Type of investment
investor_count	Number of Investors

Column Name	Description
investor_name	Name of the investor
name_investors	Name of the investor connected with the investment
investor_type	Type of entity
investor_types	Type of investor, e.g., angel vs private equity vs venture capital
investor_uid	ID used to identify investor
is_lead_investor;;	Field denoting whether or not an investor is a lead investor in the investment
last_funding_on	Date of the company's most recent funding round
lead_investor_uuids	ID(s) used to identify the lead investors involved in the funding round.
legal_name	Legal name of the organization
linkedin_url_investors	LinkedIn URL of Investor
linkedin_url_organization	LinkedIn URL of Organization
logo_url_investors	URL to the logo of the investor
logo_url_organization	URL for the image related to the organization
name_funding	Name of the funding round
name_investors	Name of the investor
name_organization	Name of the organization
num_exits	The sum of all exits related to the organization
num_funding_rounds	The sum of all funding rounds associated with the organization
org_name	Name of the organization
org_uuid	ID used to identify the organization
permalink_funding	ID used to identify the funding round
permalink_investors	ID used to identify the investor
permalink_organization	ID used to identify the organization
phone	Phone number of the company
post_money_valuation	Post money valuation attributed to the organization after the funding round.
post_money_valuation_currency_code	Currency code of the post-money valuation attributed to the organization after the funding round
post_money_valuation_usd	Post-money valuation was attributed to the organization after the funding round in USD.
postal_code	Postal Code of Company's HQ
primary_role	Role of company
raised_amount	Money raised in the funding round
raised_amount_currency_code	Currency code of the funding round
raised_amount_usd	Money raised in the funding round in USD
rank_funding	Algorithmic rank assigned to the top 100,000 most active funding rounds
rank_investment	Rank of the investment related to an investment partner (empty)
rank_investors	Rank of Investor against all investors
rank_organization	Algorithmic rank assigned to the top 100,000 most active organizations
region_funding	The region associated with an organization related to the funding round
region_investors	Region investors are in

Column Name	Description
region_organization	The geographical region that the company falls into
roles_investors	Roles associated to investors, e.g., investment firms, companies, the individual investor
roles_organization	Roles of organization
short_description	A short description of the organization
state_code_funding	State code associated with an organization related to the funding round
state_code_investors	State code of the investor
state_code_organization	State code of the organization
status_organization	Status of an organization
total_funding_currency_code_investors	Currency code of total funds raised
total_funding_currency_code_organization	Currency code of total funding amount raised across all funding rounds
total_funding_investors	Total funds raised - dollar amount.
total_funding_organization	Total funding amount raised across all funding rounds
total_funding_usd_investors	Total funds raised - dollar amount.
total_funding_usd_organization	Role of company
twitter_url_investors	URL of investor's Twitter
twitter_url_organization	URL of the organization's Twitter
type_funding	Type of entity
type_investors	Type of entity
type_organization	Type of entity
updated_at_funding	Date when information about the funding round was updated
updated_at_investment	Date when information about investment was updated on Crunchbase
updated_at_investors	Date when investor information was updated on Crunchbase
updated_at_organization	Date when company information was updated on Crunchbase
uuid_funding	ID used to identify funding round
uuid_investors	ID used to identify investor
uuid_organization	ID used to identify the organization
address_organization	Organization's street address
announced_on	Date the funding round was announced.
name_organization	name of the organization related to the category
permalink_investment	ID used to identify the investment
uuid_investment	ID used to identify the investment

19. Appendix B

19.1.1. Default Values of Algorithms in Python

Parameter name	Simple tree	Random forest	GBR
n_estimator	x	100	100
max_features	None	1.0	None
max_depth	None	none	3
min_samples_split	2	2	2
min_samples_leaf	1	1	1
learning_rate	x	x	0.1

For more information about hyperparameters, see Resources in the official documentation.

19.1.2. Values of Hyperparameters Examined by Grid Search

Parameter name	Simple tree	Random forest	GBR
n_estimator	x	100,300,500	100, 300,500,600
max_features	x	sqrt, log2	None
max_depth	3, 5, 10, None	10,30, None	3,5,7, 10
min_samples_split	2,5,10	2,10	2,5, 10
min_samples_leaf	1,2,4	1,4	1, 2, 4
learning_rate	x	x	0.1, 0.2, 0.01, 0.04

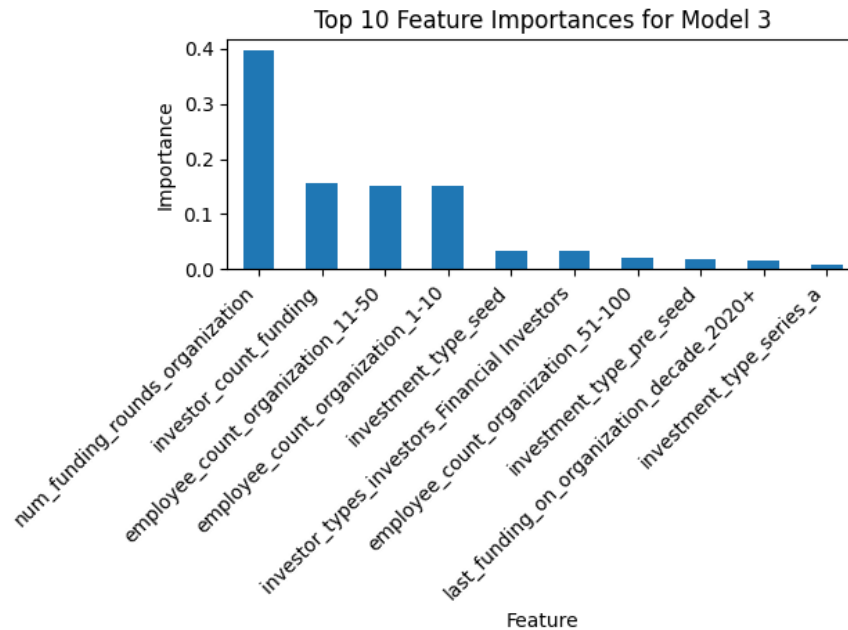
20. Appendix C

20.1. The Remaining Figures not Shown in the Main Body

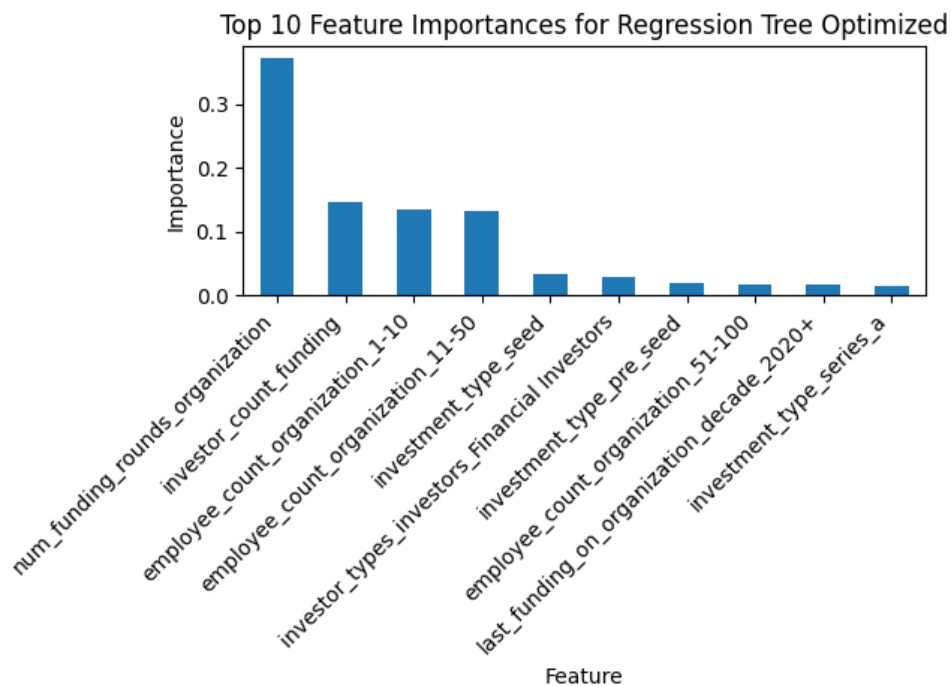
- All Figures are based on data from Crunchbase

20.1.1. Feature Importance

20.1.1.1. Regression Trees

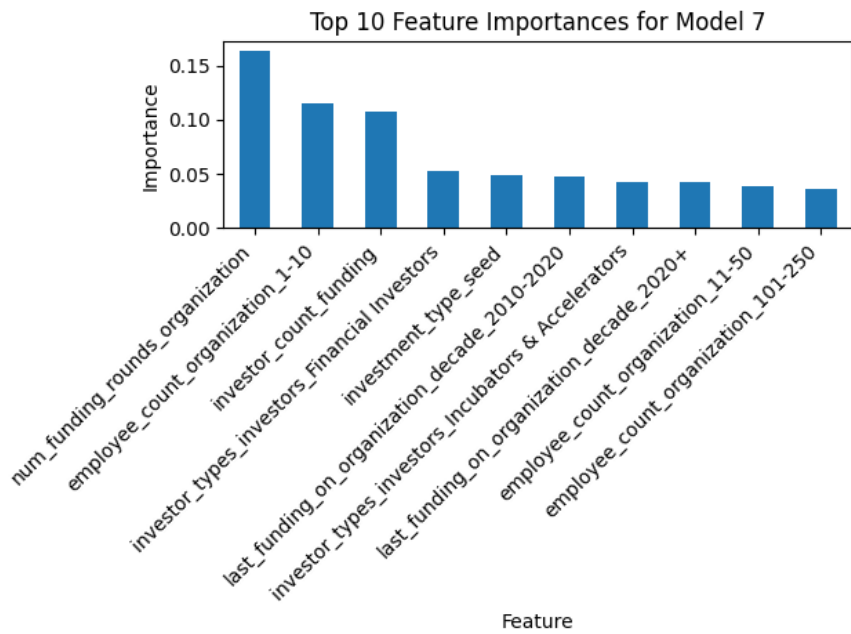


Source: Based on data from Crunchbase

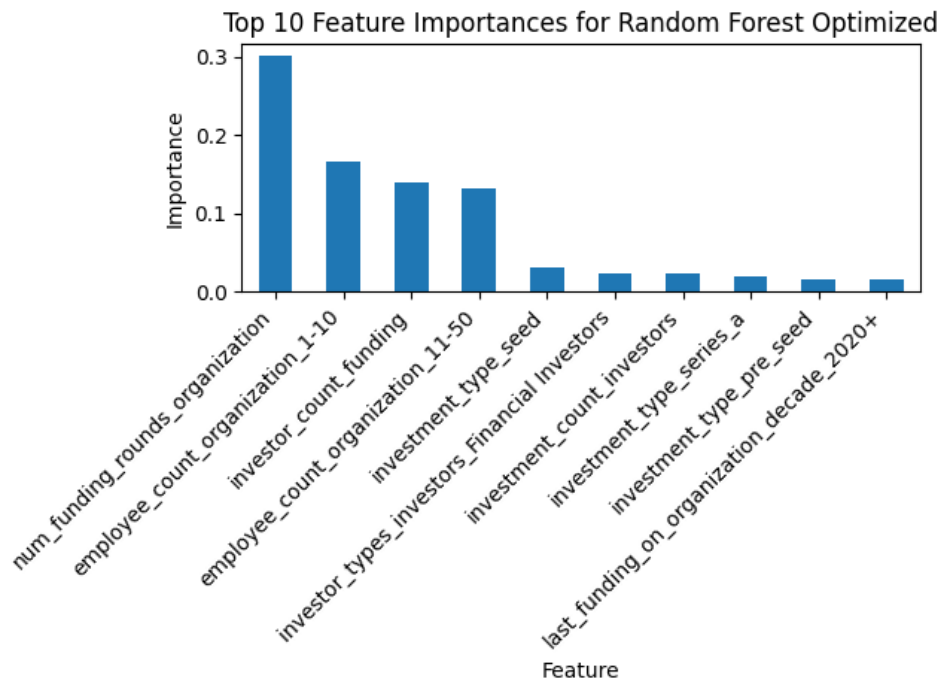


Source: Based on data from Crunchbase

20.1.1.2. Random Forest

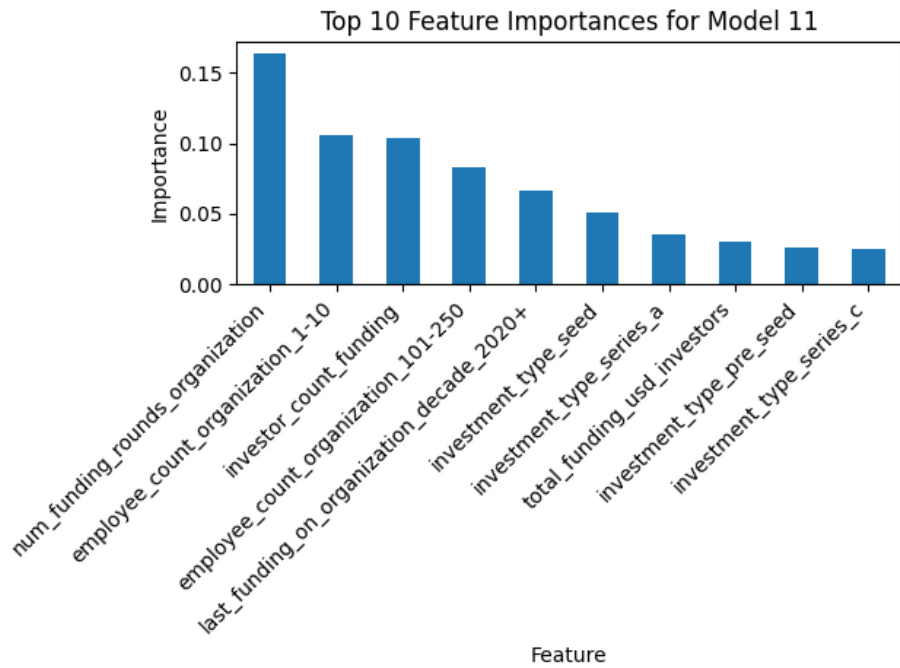


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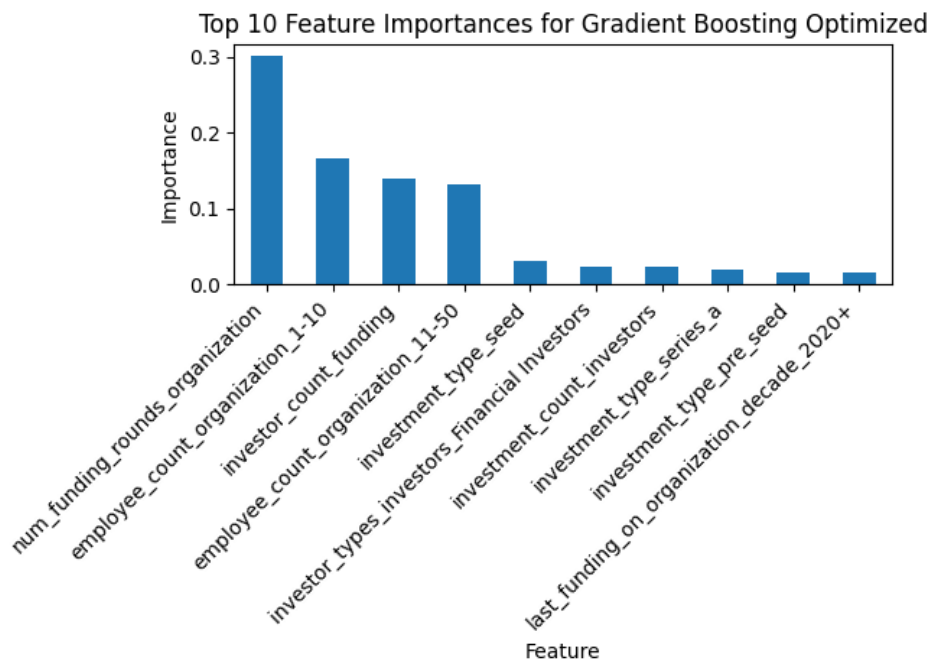


Source: Based on data from Crunchbase

20.1.1.3. Gradient Boosting



Source: Based on data from Crunchbase



Source: Based on data from Crunchbase

21. Appendix D

21.1. The Modified Models Results

- All Tables are based on data from Crunchbase

21.1.1. Regression Tree

Metrics Train	Model 2	Model 3	Model 4
MSE	1.97	1.41	0.82
MAE	1.07	0.90	0.68
RMSE	1.40	1.19	0.91
R ²	0.63	0.737	0.85

Metrics Test	Model 2	Model 3	Model 4
MSE	2.01	1.47	1.17
MAE	1.09	0.93	0.81
RMSE	1.42	1.21	1.08
R ²	0.65	0.743	0.80

21.1.2. Random Forest

Metrics Train	Model 6	Model 7	Model 8
MSE	0.992	0.87	1.09
MAE	0.77	0.71	0.80
RMSE	0.996	0.93	1.04
R ²	0.82	0.84	0.80

Metrics Test	Model 6	Model 7	Model 8
MSE	1.21	1.16	1.28
MAE	0.80	0.82	0.87
RMSE	1.10	1.08	1.13
R ²	0.79	0.80	0.78

21.1.3. Gradient Boosting

Metrics Train	Model 10	Model 11	Model 12
MSE	3.04	0.41	0.37
MAE	1.37	0.44	0.43
RMSE	1.74	0.64	0.61
R ²	0.433	0.923	0.929

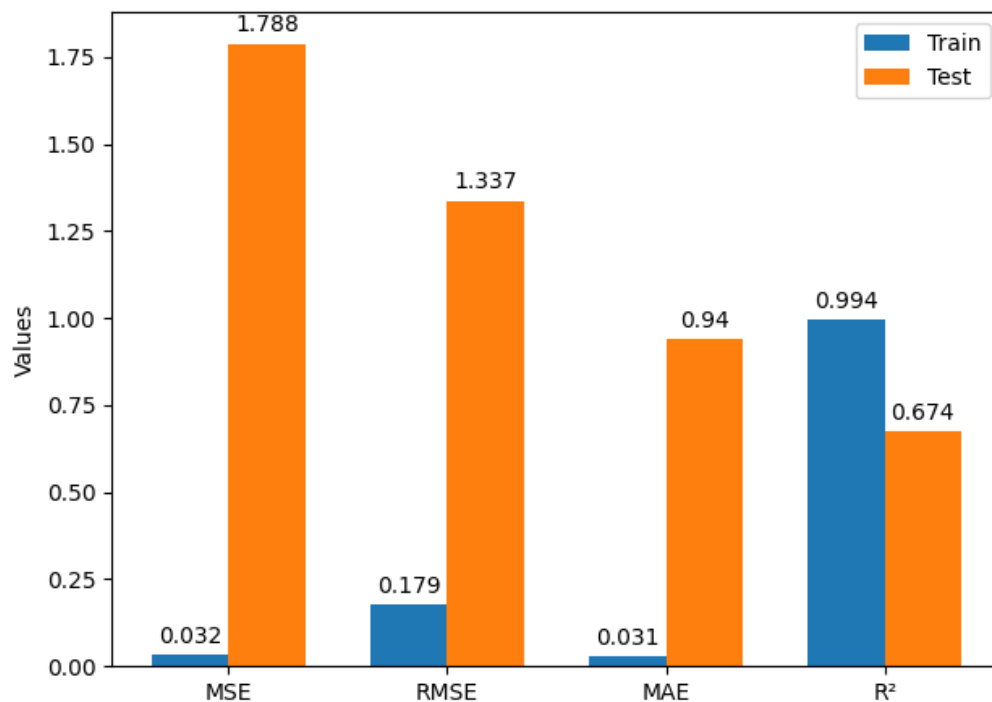
Metrics Test	Model 10	Model 11	Model 12
MSE	3.28	0.97	0.89
MAE	1.43	0.74	0.71
RMSE	1.81	0.98	0.94
R ²	0.428	0.83	0.85

22. Appendix E

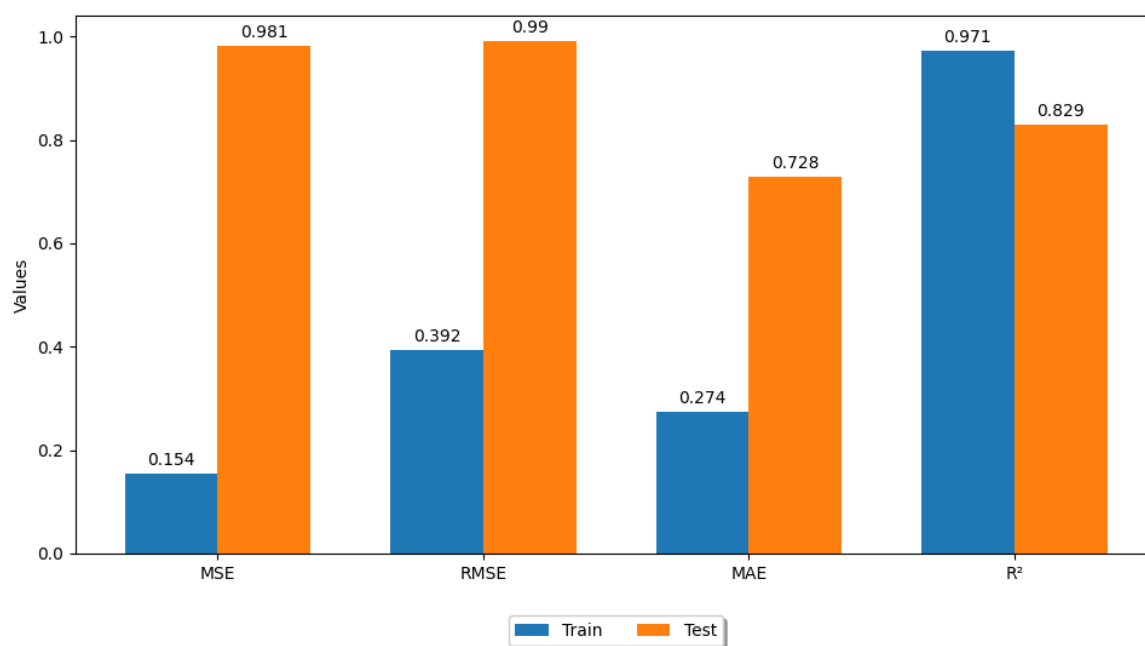
22.1. Results of the Basic Algorithms

- All Figures are based on data from Crunchbase

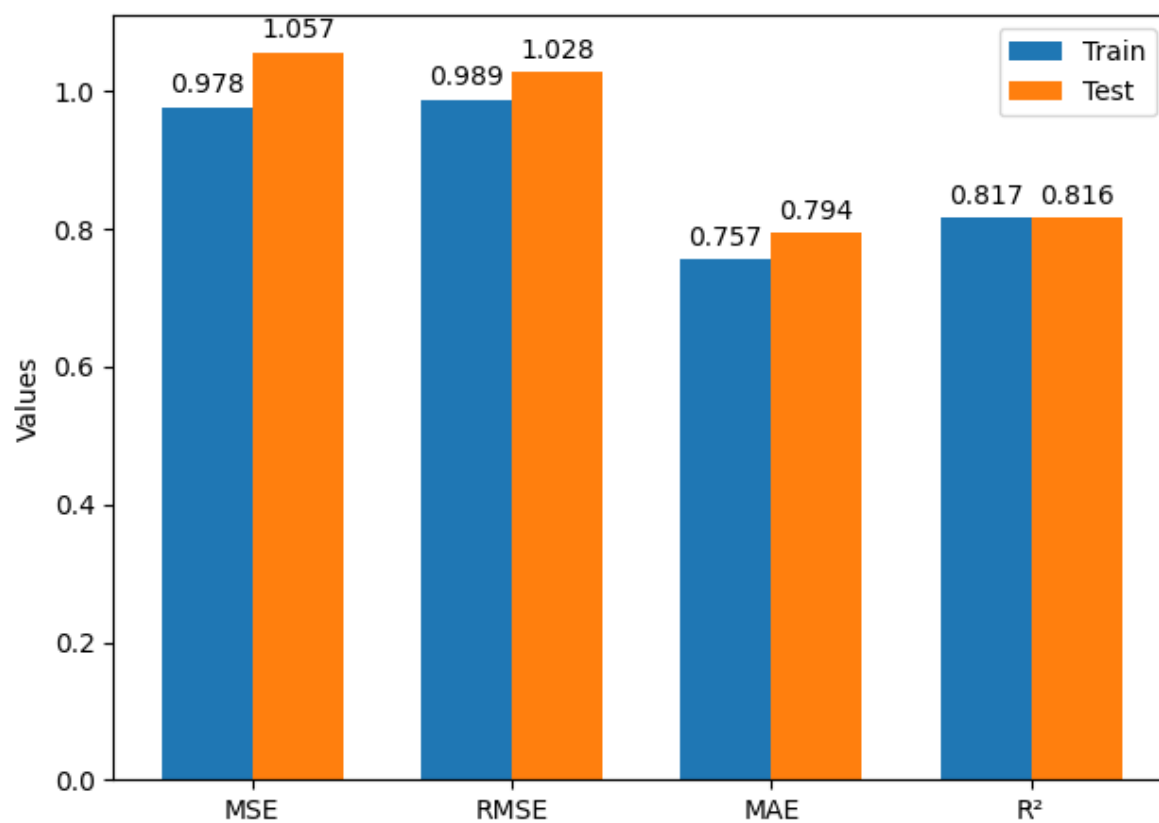
22.1.1. Regression Tree Basic Model



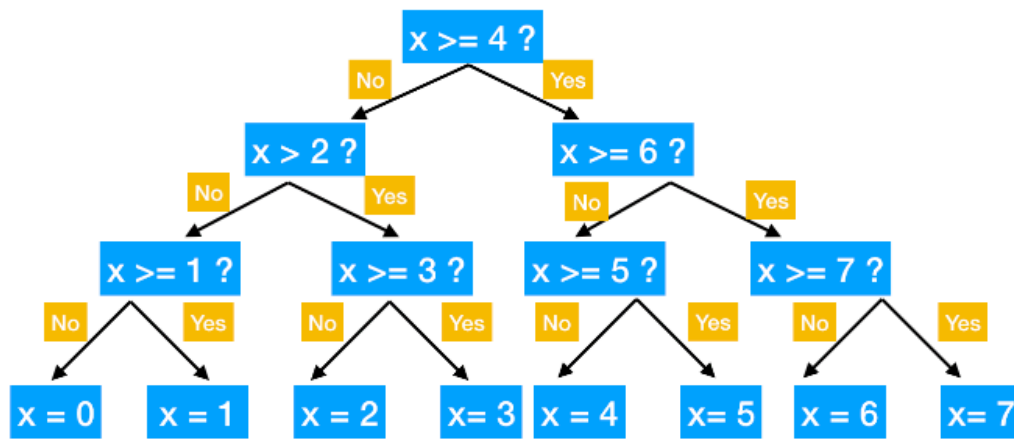
22.1.2. Random Forest Basic Model



22.1.3. Gradient Boosting Basic Model



22.2. Sample Visualization of a Regression Decision Tree



Source:(Li, n.d.)